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Master's Degree in
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**Tuscan Wine in a Changing Climate: The Impact of
Extreme Weather Events on Quality and Price**

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Dedication

To my family and lifelong friends,
for their unwavering love and support.

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Abstract

This thesis investigates the impact of climate change—specifically shifting temperature baselines and extreme weather events—on the expert-rated quality and market price of Tuscan wines. Given the economic importance and climatic sensitivity of Mediterranean viticulture, the study integrates agronomic climate indicators with hedonic pricing methods. Using a high-dimensional fixed-effects panel dataset covering 15 vintages (2004–2018), the analysis combines 1,079 expert wine ratings and real release prices from Wine Spectator with downscaled meteorological data and bioclimatic indices, including the Huglin Index and growing season temperature. The empirical strategy employs econometric specifications designed to distinguish between cumulative heat exposure and acute thermal shocks.

The results indicate an inverted U-shaped relationship between cumulative growing season temperature and both wine quality and price, with an estimated optimal threshold of approximately 20°C. Beyond this threshold, additional heat exposure is associated with declining outcomes. Extreme heat events also exhibit differentiated effects: late-season (August) heatwaves are associated with reductions in expert ratings during the *véraison* period, while total-season heatwave exposure is associated with lower market prices.

The analysis further highlights spatial heterogeneity within Tuscany. Inland areas exhibit stronger negative responses to thermal stress in both quality and price models, whereas coastal areas show weaker or statistically insignificant effects, consistent with a moderating maritime influence. With respect to regulatory frameworks, PDO classifications are associated with more stable quality outcomes relative to PGI wines under extreme heat conditions; however, no corresponding mitigation is observed in price responses.

Overall, the findings suggest that climate-related risks affect both perceived wine quality and market valuation, with implications for the economic resilience of traditional appellation systems and the need for geographically differentiated adaptation strategies.

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List of Acronyms

kha	Thousand Hectares
khl	Thousand Hectolitres
mhl	Million Hectolitres
EU	The European Union
OIV	International Organisation of Vine and Wine (Internationale de la vigne et du vin), an intergovernmental organisation.
GI	Geographical Indications
PDO	Protected Designation of Origin
DOC	Denominazione di Origine Controllata, the traditional Italian designation corresponding to the EU PDO category.
DOCG	Denominazione di Origine Controllata e Garantita, the highest Italian designation within the EU PDO category, with stricter regulations and quality guarantees.
PGI	Protected Geographical Indication
IGT	Indicazione Geografica Tipica, the traditional Italian term for a type of wine that has a PGI, interchangeable with PGI within the Italian context.
CLC	Corine Land Cover data product, updated with new status and change layers every six years.
GST	Growing Season Temperature
GSP	Growing Season Precipitation
HW	Heatwaves
FE	Fixed Effects

Chapter 1: Introduction

1.1 Studying Wine and Climate

Wine is often chosen as an excellent study subject for understanding climate change impacts on agriculture. The production, science, and economy of wine are a unique combination of high climate sensitivity (Van Leeuwen et al., 2024), well-documented historical records, and economic importance. The study of wine and climate has a multifaceted nature that provides insights applicable to broader agricultural systems (Mindlin et al., 2024). Our study begins with this aspect: wine production's sensitivity to climate change, and then examines the tangible impacts of the weather phenomena associated with this change, e.g., heat waves and changing precipitation, over a 15-year period from 2004 to 2018.

Climate change has direct impacts on wine production. The increasing and higher temperatures can lead to advanced maturation of wine grapes (Van Leeuwen et al., 2024). The changes in precipitation patterns and water availability can disrupt grape development and impact yield and quality (Di Carlo et al., 2019). More frequent severe weather events, such as hail, frost, droughts, and heat waves during the growing season can cause direct damage to vineyards (Fraga et al., 2013). These direct impacts of climate change lead to fluctuations in both yield and wine quality, as climatic variables can alter the grape's sugar content, acidity, and aroma profiles (Cassago et al., 2025).

It is evident that the viticulture in Mediterranean climate regions (such as Tuscany in Italy, Provence in France, Napa Valley in California USA, Maipo Valley in Chile, etc.) is sensitive and vulnerable to climate change, as these regions are getting warmer at a higher rate (20% more) than the global average (Cramer et al., 2021). It is also predicted that the overall precipitation in these regions is decreasing by 10% ~ 15% in the next decades, with a higher frequency of extreme events, e.g., higher intensity storms, wildfires, extended drought periods, and anomalies in rainfall patterns (Cramer et al., 2021). These shifts in climate pose risks to major premium viticultural regions.

1.2 Italian Wine and Tuscany

Italy is one of the world's largest wine producing countries. Climate change poses significant risks to its multi-billion-euro wine industry. According to the International Organisation of Vine and Wine (OIV) data, Italy produced 44.1 mhl of wine and was the largest wine producing country in 2024, accounting for nearly 20% of the world's total wine production that year. It is also the only country reporting positive growth in terms of vineyard surface area among the top 7 countries, reaching 728 kha as the No.4 in the world. By wine consumption, Italy is the second-largest market (22.3 mhl) in the EU and third globally. By international trade, Italy is the world's largest wine exporter both by volume (21.7 mhl) and by value (€8.1 billion).

Climate change is a major contributor to the environmental footprint of Italian wine production. Key factors include energy use in wineries, glassmaking for packaging, and vineyard management practices such as planting and uprooting vines (Christ & Burritt, 2013). These activities have significant

environmental consequences at both the Italian and European levels, making it crucial to understand and mitigate their impacts (Frasnetti et al., 2024).

Focusing on Tuscany is strategic for this study because Tuscany represents one of the most climate-sensitive wine regions in Italy, combining strong coastal–continental climatic contrasts. Specifically, the balance between the continental heat of the interior (comprising provinces such as Siena and Florence) and the Mediterranean influence of the coast (encompassing Grosseto and Livorno) creates an ideal environment for studying thermal stress.

This geographical dichotomy provides the necessary spatial variation to econometrically isolate the heterogeneous vulnerabilities of viticultural sub-regions to shifting bioclimatic indices. Inland macro-terroirs exhibit higher heat accumulation, exposing grapevines to severe thermal shocks during the critical ripening windows. Our high-dimensional fixed-effects modelling indicates that these interior zones experience substantial economic and qualitative negative impacts from extreme weather. Specifically, exposure to 99th-percentile August heatwaves is associated with a statistically significant degradation in expert quality scores and a non-linear devaluation in wine market valuation.

Conversely, the coastal provinces offer a critical comparative baseline demonstrating a coastal moderating effect. The maritime influence appears to buffer these vineyards against acute August thermal stress, mitigating both the physical quality degradation and associated price reductions. By exploiting Tuscany’s distinct coastal-continental contrast, this thesis can rigorously bridge climate science and hedonic pricing, utilising the region as an optimal case study to provide empirical evidence that global wine markets substantially, but heterogeneously penalise climate-induced stress.

Tuscany is historically recognised for producing high-quality wines that are globally renowned and strongly convey territorial identity. The region's wines have maintained high demand, even during periods of overall decline in Italian wine production, thanks to their distinctive regional brands and competitive advantage (von Münchhausen et al., 2023).

Renowned for its iconic wines such as Chianti Classico, Brunello di Montalcino, and Super Tuscans, the region of Tuscany is a major producer of high-quality and high-value wine production. Tuscany has in total 58 Geographical Indications (GIs) across 3 classification levels, which include 11 DOCG, 41 DOC, and 6 IGT designations. GIs were introduced in the European Union in 1992 and were later updated in 2006 and in 2012 (European Union, 1992). The GI system has proven successful in addressing market failures arising from information asymmetry, in a setting characterised by a large number of competitive producers (Menapace & Moschini, 2024). Geographical Indications (GIs) also face unprecedented challenges from climate change. As a result, tensions between tradition and innovation emerge in the process. The GIs framework, as a mechanism of quality provision for the buyers (and the producers too), depends on the stable link between place, production methods, and quality. If, hypothetically, extreme weather events (floods, droughts, heatwaves) are to affect the quality of the final bottle of wine negatively, consumer trust in the GI label is undermined. Mediterranean climates, such as those in Tuscany, California, and Southern Europe, are experiencing measurable adverse impacts to viticulture due to climate change. This includes altered timing of grape development, heat stress, reduced yields, and changes in wine characteristics, all of which are relevant to Tuscany’s situation and can have substantial economic consequences for wine producers (Graveline & Grémont, 2021; Laroche-Pinel et al., 2024).

The region of Tuscany is a driving force in Italian wine exports, playing a crucial role for the country’s international wine trade. The Italian National Statistical System (SISTAN) data show that in 2024, the Tuscany region holds more than 15% of national exports of wine (Table 1.1, row 2). Between 2023 and 2024, the value of Tuscany's wine exports grew by 8.7%, to about 1248 million euros, outpacing the national average growth rate of 5.5% for wine exports. Tuscany is among the leading regions in this sector. Veneto, Tuscany, and Piedmont account for 66.7% of Italian wine exports in 2024.

Table 1.1: Wine Export per Region in the Year 2024 in Italy Ranked by Export Value (in 1000 €).

	Region	Export Value (1000 €)	National Share	YoY Growth*
1	Veneto	2,996	36.8%	7.3%
2	Tuscany	1,248	15.3%	8.7%
3	Piedmont	1,184	14.5%	0.1%
4	Trentino-Alto Adige	611	7.5%	−2.8%
5	Emilia-Romagna	465	5.7%	0.6%
6	Lombardy	313	3.8%	−2.8%
7	Abruzzo	269	3.3%	16.8%
8	Friuli-Venezia Giulia	226	2.8%	9.0%
9	Apulia	232	2.8%	9.1%
10	Sicily	155	1.9%	−2.5%
	...	...		...
	Italy	8138	100.0%	5.5%

Note: YoY growth: Compared to the year of 2023. Data source: Regione del Veneto (2025).

In summary, Tuscany is important in climate-wine research because of its economic importance, worldwide prestige wine reputation, and the significant impacts it experiences from climate change. This study can contribute to region-level studies on how Tuscan wines respond to climate change. Our research question is: How do specific weather variables (temperature, precipitation, extreme weather events) affect the quality and price of Tuscan wines? The objective of this study is to quantify the impact of weather on wine quality and price to determine the potential economic consequences caused by climate change, specifically its effect on undermining the quality signals traditionally associated with specific geographical indications.

1.3 Thesis Structure

To answer our research question, we constructed a comprehensive panel dataset of 1079 observations about Tuscan wine which consists of expert ratings by Wine Spectator (WS) magazine with vintage from 2004 to 2018, winery information, and release prices. We then utilise Google Maps API to locate each winery or vineyard’s locality, matching with the official ISTAT codes. With the wine names, we mapped each rated wine with the EU Geographical Indications for food and wine classification such as PDO or PGI. We then employed a panel fixed-effects model to control for time-invariant unobservables and isolate the effect of weather fluctuations.

By combining high-resolution, vineyard-weighted climatological indicators—such as the Huglin Index and heatwave metrics—with expert sensory evaluations and normalised market prices, this study develops a quantitative framework for analysing climate impacts on wine outcomes. This empirical approach links agronomic climate impact analysis with hedonic pricing models.

The framework enables an assessment of both the physiological response of grapevines to environmental stressors across different macro-terroirs and the extent to which these effects are reflected in market prices. The following chapters present the empirical implementation of this approach.

This thesis consists of 6 chapters. After Chapter 1 introduces the reasoning and motivation behind the research topic, Chapter 2 gives a comprehensive literature review of related wine-climate research, including some of the most foundational studies, concepts, and theories such as Dr. G.V. Jones' work. Chapter 3 describes the data sources, data collection, data processing, and the definitions of the variables. Chapter 4 lays out the econometric modelling: a dual analysis of quality and price impacts. Chapter 5 discusses the results and Chapter 6 draws the conclusions.

Chapter 2: Literature Review

The study of weather variation's effects on wine quality is supported by both recent research (2020-2025) and several foundational papers that have shaped scientific understanding and methodologies in this field.

2.1 Terroir Theory and Climate as a Determinant of Wine Quality

Jones et al. (2005) in their landmark work established systematic links between climate variables (temperature, precipitation, degree days) and wine quality ratings across regions and vintages. The concept of "climate maturity groupings" for grape varieties based on growing season temperature remains a pillar in wine-climate research, widely used to predict which regions are optimal for specific grapes and how those regions might shift under climate change. Jones' research quantified how weather variability relates to grape ripening, phenology, and the sensory profiles of wines.

Van Leeuwen & Seguin (2006) defined terroir as a multidimensional construct, integrating climate, soil, grapevine variety, vineyard management, and microbial communities into a holistic framework for understanding wine quality. Wine and climate are intrinsically linked. In the wine industry and wine research literature, the importance of terroir and climate in determining the quality of the final bottle is well known. Temperature and precipitation are among the most critical climate factors in creating a favourable terroir.

2.2 Focus of Climate-Wine Research in the Literature

Based on a review of the literature, we can find that the primary focus of climate-wine research is on the effects on wine grapes and production, and that there is limited focus on how climate affects expert ratings.

2.2.1 Extensive Research on Grapes and Viticulture

Many studies have focused on the effects of climate on grape quality, grapevine physiology, and the resulting wine's chemical composition. This includes research on how temperature, sunlight, soil, water availability, and other environmental factors (collectively known as "terroir") influence grape and wine quality. Climate is often considered the main constraining factor, and the phenolic composition of wines has been extensively studied in relation to climate variables (Li et al., 2021; Nicholas et al., 2011).

There is a strong emphasis on understanding how climate change impacts grape-growing conditions, grapevine development, and the overall suitability and sustainability of wine production regions (Van Leeuwen et al., 2024; del Río et al., 2024; Christ & Burritt, 2013; Crouchett-Rojas et al., 2025). As climate change continues to reshape Italian viticulture, its direct biological impacts are becoming increasingly evident. Long-term historical analyses demonstrate that shifting temperature baselines over recent decades have profoundly altered grapevine phenology and overall wine quality across Italy (Faralli et al., 2024). To adapt to these accelerated developmental cycles and the rising threat of extreme weather events, the viticultural sector is increasingly looking toward data-driven solutions. The integration of advanced climate services and weather forecasting tools into daily vineyard

management has proven highly valuable, providing producers with the predictive data necessary to mitigate climate risks and make informed agronomical decisions in a rapidly changing environment (Khosravi et al., 2024).

2.2.2 Expert Ratings and Climate: Gap in Literature

While wine experts are aware of the influence of climate and terroir on wine characteristics, their primary focus is on sensory evaluation (taste, aroma, appearance, etc.) rather than directly studying the impact of climate on expert ratings themselves (Joy et al., 2020).

The literature does mention "expert opinion" as a research topic in wine economics, but there is no clear indication that many studies specifically analyse how climate affects expert ratings of wine, as opposed to grape or wine quality in general (Caldas & Rebelo, 2013).

Most research cited is concerned with the physical and chemical impacts of climate on grapes and wine, not on how these changes translate into expert scores or ratings. The majority of climate-wine research focuses on the effect of climate on wine grapes and production, with very few studies directly addressing the effect on expert ratings of wine. However, there are emerging studies that explore the relationship between climate change and some specific varieties in world-class wine producing regions such as Chile's Maipo Valley (Charlin & Cifuentes, 2025).

2.3 Temperature

Some studies use mean temperature (daily, monthly, or bi-monthly average, etc.). Others like Charlin and Cifuentes (2025) incorporate extreme temperatures not just average in their models. One recent study on California wine by Whitnall & Alston (2025) used the concept of degree-day. In this section, we discuss the use of degree-day and compare it to the use of growing season temperature during growing seasons as well as the use of heat waves and extreme hot days in our study.

2.3.1 Degree-day vs. Growing Season Temperature (GST)

A degree-day is a unit of measurement that quantifies the accumulation of heat over time. It is used to estimate the growth and development of organisms like plants or insects, as temperature is a primary driver of their life cycles. In the Whitnall & Alston (2025) paper, the authors use a sophisticated version of the degree-day concept. Instead of having just one temperature interval for growing heat, they use several different temperature intervals to measure exposure to different temperature ranges.

The study defines degree-days as a measure of "for how long and by how much temperatures exceed... the lower bound of a temperature interval... without exceeding... an upper bound" (Whitnall & Alston, 2025). They calculated these values by estimating how many hours were spent within specific 1°C temperature intervals each day.

For their analysis, they created variables to measure the accumulation of degree-days in different temperature intervals:

- Freezing days: Temperatures < -2°C
- Cool days: Temperatures between -2°C and 10°C

- Hot days: Temperatures between 30°C and 35°C
- Extremely hot days: Temperatures > 35°C

This approach is advantageous because it allows them to distinguish between the effects of beneficial warmth and the harmful effects of extreme heat, something a simple average temperature cannot do.

2.3.2 Methodological Contribution

Our approach to thermal variable selection is inspired by this degree-day framework which recognises that relying exclusively upon simple, continuous seasonal averages fails to capture the true physiological realities of viticulture. Simple averages may obscure significant biological stress (Whitnall & Alston, 2025). Our structural methodological approach mirrors this highly effective analytical bifurcated logic. Consider the Huglin Heliothermic Index. By explicitly measuring the cumulative heat available to grapevines above a baseline of 10 degrees Celsius, this specific metric effectively functions as a highly targeted degree-day calculation that captures optimal fruit maturity potential.

Furthermore, our deliberate integration of specific, absolute temperature thresholds, which count the number of days surpassing 32, 33, 34, and 35 degrees Celsius, directly emulates the "hot days" temperature interval utilised in modern degree-day literature. We isolate these critical thermal thresholds. By structurally separating our baseline environmental topology from these acute extreme heat counters, we successfully achieve a comparable econometric goal to that of the degree-day temperature interval framework, which is to distinguish physiological heat stress from standard agricultural ripening.

Our contribution is adapting this logic into a novel "Comprehensive Model" structure tailored for Italian viticulture. While previous literature often relies on simple seasonal averages, our contribution lies in demonstrating that combining cumulative baseline heat indices (Huglin Index / GST) with acute extreme threshold counters (99th-percentile heatwaves and isolated days above 32-35°C) significantly improves the model's explanatory power (e.g., moving from simple models to comprehensive models improved the Within R-squared from approximately 7.6% to over 12%).

2.3.3 August Heatwaves

Heatwaves occurring in August can cause significant damage to both the physical condition of the grapes and their delicate chemical balance such as significant reductions in acidity (Micciché et al., 2025) and loss of pigment and aroma (Faralli et al., 2024). Heatwaves often bring excessively warm nights alongside extreme daytime temperatures. High nighttime temperatures in August disrupt the plant's nighttime respiration, impair the repair of damaged photosystems, and interfere with carbohydrate translocation (Rienth et al., 2016; Tombesi et al., 2019; Crupi et al., 2024). This lack of recovery reduces the vine's photosynthetic efficiency for the following day. Ultimately, heatwaves compromise the harvest and the resulting wine quality. Because extreme heat abnormally accelerates the vine's phenological development, grapes are forced to mature under warmer conditions in a much shorter timeframe (Micciché et al., 2025). This frequently necessitates a compressed, much earlier harvest period to prevent the grapes from over-ripening or severe dehydration, making it highly challenging to maintain the traditional characteristics and typicity of the wine (Faralli et al., 2024).

The empirical results of this study regarding extreme thermal events in August align closely with established viticultural research. Our statistical analysis reveals that an increase in the number of August heatwave days exerts a significant negative pressure on both wine quality scores and market prices. Specifically, models utilising comune-level fixed effects indicate that each additional extreme hot day in August is associated with a significant decrease in price and quality perception.

2.4 Precipitation

Growing Season Precipitation (GSP) is the most appropriate hydrological metric for this study because, first, GSP is a critical meteorological descriptor and its use is well-established in viticultural climate literature (Jones & Goodrich, 2007); and second, all the wines in our dataset are either PDO or PGI and there are regulations on irrigation in Italy. Within the Italian viticultural framework, the regulatory landscape renders GSP a critical, exogenous econometric variable rather than a mere meteorological descriptor (Commission Regulation No 753/2002, 2002). The dataset utilised in this study is overwhelmingly comprised of wines holding European Geographical Indications, specifically PDO (54.77%) and PGI (45.13%) classifications. These designations impose strict agronomical constraints that fundamentally alter how vineyards interact with the climate (Menapace & Moschini, 2024).

Most crucially, the production regulations for high-tier PDO (DOC/DOCG) wines enforce strict dry-farming paradigms, mandating prohibitions on routine irrigation and only permitting artificial watering under extreme, officially declared emergency conditions. While PGI (IGT) regulations offer marginally broader flexibility for irrigation, the regional standard across premium Tuscan viticulture remains overwhelmingly reliant on natural rainfall.

Because producers are legally constrained from artificially compensating for water deficits to manipulate yields or canopy health, GSP (April–September) operates as a proxy for the natural water stress the vines endure during the critical berry development and ripening (*véraison*) windows. By utilising GSP, our econometric specification captures how this rigid regulatory framework affects Tuscan wineries, making them tightly bound to precipitation conditions evolving across vintages. The results of this study suggest that global markets may increasingly factor regional water availability into wine release prices.

Nevertheless, this cumulative approach has inherent limitations. As highlighted by Dunn et al. (2019) in their study of high-latitude viticulture, utilising broad precipitation averages can obscure critical seasonal variations and extreme moisture events that trigger fungal disease or physical fruit damage. Furthermore, unlike suitability models that integrate high-resolution lithology and topography to assess meso-climatic resilience, this study's reliance on municipality-level cumulative flux may under-represent the impact of specific high-intensity storm events during the harvest window. Future research could benefit from disaggregating GSP into monthly increments to better isolate the timing-specific sensitivity of Tuscan wine prices to late-season hydrological shocks.

2.5 Wine Ratings and Wine Prices

There is substantial evidence supporting a link between perceived wine quality and wine price. A study of French wines finds that sensory characteristics are more strongly correlated with price than

objective traits (Lecocq & Visser, 2006). Analyses in Australia show that the reputation and ratings of wines (which reflect perceived quality) are linked to output price, supporting the idea that perceived quality can influence wine pricing (Lock et al., 2019). Oczkowski and Doucouliagos (2015) consistently find a positive correlation between wine price and perceived quality across approximately 180 hedonic wine-price models spanning 20 years and multiple countries.

Building upon these empirical studies and experimental evidence, this study aims to explore the dynamics among weather variation, wine ratings, and wine price. Our primary empirical contribution is simultaneously quantifying the impact of acute thermal shocks on both subjective expert ratings and real market-clearing prices using a high-dimensional fixed-effects panel dataset (2004–2018). Our work is to bridge the gap between agronomic climate impact studies and hedonic wine pricing.

Chapter 3: Data and Sources

3.1 Wine Quality

We use expert ratings as a proxy for wine quality, specifically perceived wine quality. Analysing perceived quality is different from analysing the chemical composition of the wine in terms of wine quality, but in the context of climate change and agriculture and food economy, the perceived quality data is more relevant to the consumer experience.

Expert ratings are typically based on objective, descriptive attributes of the wine. Most importantly, they are often quantified, which makes them ideal for economic modelling and for our aim to explore the quantitative impact of climate on wine quality.

We sourced the expert ratings data from Wine Spectator, which included the name of the winery, name of the wine, vintage, rating score, and the wine’s release price in USD. Wines are tasted blind by Wine Spectator expert tasters and the rating scores are on a 50-100-point scale. A score above 80 is considered good (Table 3.1).

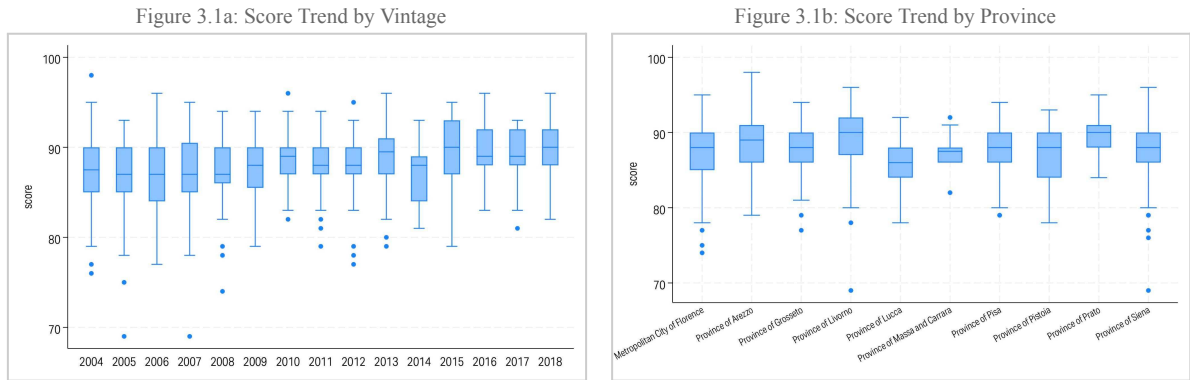
Table 3.1 Wine Spectator’s 50-100-Point Scale

Score	Wine Quality Classification
95-100	Classic: a great wine
90-94	Outstanding: a wine of superior character and style
85-89	Very good: a wine with special qualities
80-84	Good: a solid well-made wine
75-79	Mediocre: a drinkable wine that may have minor flaws
50-74	Not recommended

Source: Wine Spectator Magazine

We chose Wine Spectator for its large coverage. It is one of the largest wine databases with more than 440,000 ratings, covering 18 countries and more than 100 wine producing regions (and sub-regions), categorising 33 types of wines with vintage from the 1770s to the current year (e.g., 2025). It is also the most referenced wine rating data source (among the top 3 wine expert critics platforms: Wine Spectator, Robert Parker, and Enthusiast) in wine research to the best of our knowledge as of September 2025. Google Scholar search results show that: between 2000 and 2025, over 40,100 publications cited Wine Spectator, compared to 22,800 citing Robert Parker and 16,500 citing Wine Enthusiast (Google Scholar, 2025). We acknowledge Wine Spectator’s limitation of not reviewing certain types of wine as mentioned by some studies. However, in our study, this limitation is negligible because Tuscany is a region where wineries mostly produce high-quality, prestige wines (Figure 3.1, next page).

Figure 3.1: Tuscan Wine Scores Overview



3.2 Wine Price

We sourced the price data from Wine Spectator as they have consistently labelled a price to each of their wine ratings. It is called “Release Price” and is defined as the price of the wine when it was first released to the market. Producers and retailers provide Wine Spectator with this information. When the suggested retail price is not available, a prevailing wholesale price is used instead. When a price has not been set yet, “\$NA” is used and it is then treated as a missing value in our dataset.

As the first step of processing the price data, we normalised the price to address different volumes in the dataset. All prices have been standardised to have the same measuring unit, 750ml, a standard volume for a regular bottle of wine.

As a second step, we addressed the comparison of prices across different time periods. We used the annual average Consumer Price Index (CPI) as the deflator to transform the nominal price of each wine from its respective vintage year into a real price. We applied the formula below:

$$Real\ Price = Nominal\ Price \times \left(\frac{CPI_{base\ year}}{CPI_{observation\ year}} \right) \quad (3.1)$$

The year 2018 was selected as the base year for this study. Thus all resulting real prices are now expressed in terms of 2018 USD. By adjusting all the nominal prices in this analysis for inflation, we removed the distorting effects of general price level changes over time.

3.3 Climatological Variables and Bioclimatic Indices

3.3.1 Fundamental Climate Data

The fundamental weather data (daily mean temperature, daily maximum temperature, and daily precipitation) is sourced from the AgERA5 dataset: Agrometeorological indicators derived from ECMWF Re-Analysis (5th generation), provided by the Copernicus Climate Change Service.

The raw AgERA5 gridded data is then downscaled to the specific boundaries of Italian municipalities (comune) using ISTAT shapefile. Moreover, the CORINE Land Cover 2018 dataset (100 m resolution) is used to identify the area covered by vineyards within each municipality. This information is then used to create a vineyard-weighted average of the weather variables. The purpose is to refine the downscaling from a simple geographical average to a much more precise, more representative climate indicator of the actual conditions experienced by the vineyards in the comune.

Table 3.2: Fundamental Climate Data Descriptions

Name	Units	Description
2m temperature	K	Air temperature at a height of 2 metres above the surface.
T_max	°C	24 hour maximum
T_mean	°C	24 hour mean
Precipitation flux	mm/day	Total volume of liquid water (mm ³) precipitated over the period 00h-24h local time per unit of area (mm ²), per day.

3.3.2 Bioclimatic Indices

Three key bioclimatic indices were calculated to capture the aspects of the growing season in viticulture: the Huglin Index (HI) for thermal potential, the mean temperature of the vegetative period (GST), and total growing season precipitation (GSP).

i. Huglin Heliothermic Index (HI)

The Huglin Index is a standard heliothermic index used to measure the thermal potential of a wine-growing region. It effectively captures the cumulative heat available to grapevines during the growing season (April 1st to September 30th) and is strongly correlated with grape ripening and quality. The index is calculated for each municipality using the standard formulation:

$$HI = \sum_{1 Apr}^{30 Sep} \left(\frac{T_{mean} + T_{max}}{2} - 10 \right) \times K \quad (3.2)$$

In this formula, T_{mean} and T_{max} represent the daily mean and maximum air temperatures, respectively. The value of 10°C is the baseline temperature, representing the threshold for the grapevine's active physiological cycle. The coefficient K is a latitude-based correction factor for day length, which is critical for accurately modelling heat accumulation. Based on the calculated HI values, each municipality-year is assigned a viticultural suitability class (HI Class) according to the thresholds established by Tonietto and Carbonneau (2004).

ii. Growing Season Temperature (GST)

Growing Season Temperature (GST) is calculated as the average of the daily mean temperatures from April 1st to October 31st in the Northern Hemisphere, where Italy is located. The mean temperature during the vegetative period is a critical driver of the timing of grapevine phenological stages (Massano et al., 2023). We use a 7-month period for temperature to capture the complete vegetative cycle, April–October, because it covers everything from Bud Break in early spring to Leaf Fall in late autumn (Jones et al., 2005; Nesbitt et al., 2016).

iii. Growing Season Precipitation (GSP)

Growing Season Precipitation (GSP) is a vital indicator for assessing water availability and the risk of water stress, particularly for vineyards in Tuscany Italy, where regulations for PDO wines permit only emergency irrigation, while PGI wines allow broader use of irrigation. This index is calculated as the cumulative sum of daily precipitation from April 1st to September 30th, measured in millimetres (mm).

We use a 6-month period for precipitation because it focuses on the Ripening Cycle and water stress. Besides, our calculation of GSP aligns with one of the Standard Bioclimatic Indices, Huglin Index (HI). This 6-month window is the global standard established by Tonietto and Carbonneau (2004). For wine quality, the most critical time for water availability is during berry development and ripening, i.e., *véraison* (Charlin & Cifuentes, 2025). In Tuscany, rain in October is often the start of the autumn wet season and occurs during or after the harvest. Including October rain would introduce noise into the model, as post-harvest precipitation does not biologically affect the developed grapes. Finally, we must consider the regulatory context: in Italy, PDO regulations limit irrigation. GSP (April–Sept) is used as the proxy for the natural water stress the vine faces while the grapes are actually on the vine.

Table 3.3: Calculation Periods for Bioclimatic Indices

Index	Period	Focus
GST	April to October	The lifecycle of the vine (growth to dormancy).
GSP	April to September	The development of the grapes (water stress & disease). Focus on the ripening window only to avoid harvest rain noise from October.
HI	April to September	The fruit maturity potential in a specific region.

3.3.3 Heat Indicators

i. Heatwaves (HW)

Heatwaves can induce significant stress on grapevines, impacting yield and berry composition. Following the specific climatological methodology from EURO-CORDEX for Italy, a heatwave day corresponds to a local heat warning and is triggered from day 1 when the daily maximum apparent temperature exceeds a specific high-percentile threshold between the 15th of May and the 15th of September. This threshold is calculated from the distribution of daily maximum apparent temperatures over the 1984–2003 reference period.

Two different thresholds were used to define heatwave days in this study:

- **Heatwave 99th-percentile (hw_99_3d)**: The total count of days within a given year (from May to September) that fall within a heatwave defined at the 99th percentile.
- **Heatwave 97.5th-percentile (hw_975_3d)**: The total count of days within a given year (from May to September) that fall within a heatwave defined at the 97.5th percentile.

Note: While the standard broader European variables utilise a *_3d* suffix to denote a 3-day minimum, the specific heatwave variables used in this study for Italy strictly count individual heatwave days as triggered from day 1.

These heatwave variables are also disaggregated by month (from May to September, e.g., hw_99_3d_may, hw_975_3d_jul, etc.) to analyse the specific timing of these extreme events.

ii. Hot Days (HD)

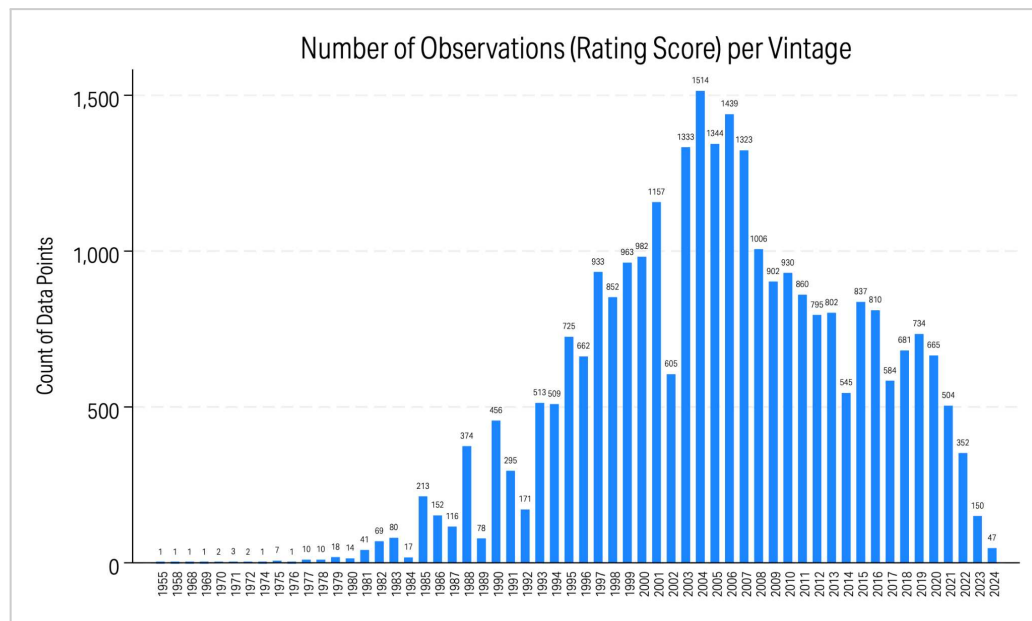
To provide a more direct measure of heat stress, we also calculated the total number of days from April to October when the maximum daily temperature exceeded specific absolute thresholds. These variables serve as straightforward counters for exposure to potentially damaging high temperatures. The variables included are:

- **Hot Days > 32°C (days_above_32)**: Count of days with $T_{max} > 32^{\circ}\text{C}$.
- **Hot Days > 33°C (days_above_33)**: Count of days with $T_{max} > 33^{\circ}\text{C}$.
- **Hot Days > 34°C (days_above_34)**: Count of days with $T_{max} > 34^{\circ}\text{C}$.
- **Hot Days > 35°C (days_above_35)**: Count of days with $T_{max} > 35^{\circ}\text{C}$.

Chapter 4: Methodology

4.1 Study Period

Figure 4.1: Number of Observations per Vintage



We have collected and utilised a comprehensive dataset of observations (wine ratings and release prices) from Wine Spectator (as shown in Figure 4.1) that allows for an in-depth analysis of a large sample size with a broader time span to address the climate shift's impact on wine ratings. However, several constraints at the time of carrying out this study were taken into consideration before the study period was set to be 2004-2018.

4.1.1 Count of Data Points

As we can see on Figure 4.1, between 1993 and 2021 (including 1993 and 2021), each and every vintage has above 500 data points (unique observations), which makes this 29-year period an excellent choice for our empirical analysis. However, this period has a considerable overlap with the temperature baseline period (1984-2003) that we chose to calculate the heat waves.

4.1.2 Heat Wave Reference Baseline

1984–2003 was used as a 20-year validation window in regional climate projections, the IPCC Fifth Assessment Report (AR5), and many other climate studies (Christensen et al., 2014; Matte et al., 2019). We chose this period as the heat wave reference baseline to ensure comparability and external validity in the scientific research community. Moreover, the modern Italian DOC/DOCG regulations were largely consolidated or matured by the 1990s (Commission Regulation No 753/2002, 2002). The chosen reference period allows for the identification of extreme events that represent significant deviations from the established climatic conditions. Thus, our study period was set to start from the year 2004 to provide a clean temporal separation from the reference baseline.

4.1.3 Land Cover Dataset Availability

To reduce the classical measurement error introduced by using comune-wide climate averages, as large shares of land are non-viticultural, we also use the Corine Land Cover (CLC) dataset from the Copernicus Land Monitoring Service to refine the downscaled climatic data for each Tuscany comune. This data treatment enhances the relevance of climate variables for wine production. Extending the analysis to include the 2019–2021 vintages would require the Corine Land Cover dataset for the 2024 reference year, which is currently unavailable (scheduled for publication in Q3 2026). Consequently, this study utilises the latest available version: CLC 2018. Although CLC 2018 is time-invariant, vineyard location in Italy is highly persistent due to regulatory and agro-climatic constraints, making it appropriate for panel estimation.

Therefore, choosing 2004–2018 as this study’s observation period is sufficient for an analysis of vulnerability and resilience in viticulture. 15 vintages constitute a significant time horizon for a winery or a region to adapt. Even though this study period does not cover recent years 2019–2021, which were among the most extreme years for Italian viticulture, this is not a limitation but rather lays the foundation for future studies that will expand to additional vintages once CLC 2024 becomes available.

4.2 Descriptive Statistics

4.2.1 Summary Statistics

The summary statistics for the primary variables, based on a final sample of 1,079 observations spanning the 2004–2018 period, are presented in Table 4.1 below.

Table 4.1: Summary Statistics of Key Variables

Variable	Obs.	Mean	Std. Dev.	Min	Max
Score	1079	87.962	3.611	69	98
Price	994	47.812	53.369	7.474	585.23
GST	1079	18.841	.978	12.57	21.32
GSP	1079	359.369	90.591	145.199	758.6
HI	1066	2286.654	178.208	1660.024	2679.906
Heatwaves [†]	1079	.45	.549	0	3
Hot Days >35°C	1079	1.939	2.829	0	16
Vintage	1079	2009.99	4.218	2004	2018

Notes: [†] Total 99th-percentile heatwaves from May to September. Obs.: Observations. Std. Dev.: Standard Deviation.

We also group the provinces based on their macro-terroir type, listed in Table 4.2 below, which can be used for segmentation analysis.

Table 4.2: Grouping of Provinces by Macro-Terroir

Macro-Terroir	Provinces	Combined N	Combined %
Coastal	Grosseto, Livorno, Pisa	380	35.21%
Inland	Florence, Siena, Arezzo	614	56.90%
Transition	Lucca, Massa, Pistoia, Prato	85	7.88%

Notes: N is “Number of observations”.

4.2.2 Correlation Analysis

Table 4.3 presents the pairwise correlation matrix for the primary variables of interest. As expected, a significant positive correlation exists between wine quality (score) and standardised price ($r = 0.448^*$, Table 4.3, column 1, row 2), suggesting that sensory quality remains a key driver of market value. Regarding climatic trends, the data reveals a significant positive correlation between the vintage year and all thermal indices, including the Huglin Index ($r = 0.295^*$, column 5) and growing season temperature ($r = 0.301^*$, column 3), confirming a warming trend over the study period. While individual climate variables show weak direct correlations with wine scores in this preliminary matrix, the high correlation between the Huglin Index and GST ($r = 0.895^*$, Table 4.3, column 3, row 5) suggests a need for careful variable selection in subsequent regression models to avoid multicollinearity.

Table 4.3: Pairwise Correlation Matrix of Quality, Price, and Bioclimatic Indices

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) Score	1.000						
(2) Price	0.448* (0.000)	1.000					
(3) GST	0.050 (0.100)	0.025 (0.431)	1.000				
(4) GSP	0.022 (0.475)	-0.014 (0.652)	-0.526* (0.000)	1.000			
(5) HI	0.016 (0.609)	-0.016 (0.617)	0.895* (0.000)	-0.572* (0.000)	1.000		
(6) Heatwaves [†]	0.054 (0.079)	-0.010 (0.755)	0.331* (0.000)	-0.339* (0.000)	0.412* (0.000)	1.000	
(7) Vintage	0.229* (0.000)	0.042 (0.187)	0.301* (0.000)	0.061* (0.046)	0.295* (0.000)	0.309* (0.000)	1.000

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † Total 99th-percentile heatwaves from May to September.

The pairwise correlation analysis reveals a distinct and statistically significant relationship between the passage of time (vintage) and nearly all climatic variables, providing empirical evidence of a warming trend in Tuscany over the study period (2004–2018). Furthermore, extreme weather events, specifically 99th percentile seasonal total heatwaves, show a significant positive correlation with the vintage year ($r = 0.309^*$, Table 4.3, column 6, row 7), indicating an increase in the frequency of high-intensity thermal stress events over time.

Interestingly, while vintage is positively correlated with most key variables, it shows no significant relationship with real prices (Table 4.3, column 2, row 7). This lack of correlation with price likely reflects the successful removal of inflationary noise through the price-deflation process ([Formula 3.1](#)), as well as the dominance of brand and winery-specific reputation over temporal trends.

These significant correlations between vintage and both climate and quality underscore a potential omitted variable bias in simple linear models; they suggest that improvements in viticultural technology or wine-making expertise may be advancing alongside rising temperatures. This finding justifies the use of the Panel Fixed Effects model described in the Econometric Models section, which is required to isolate the marginal impact of weather fluctuations from these broader temporal improvements in the industry.

Table 4.4: Pairwise Correlation Matrix of Heatwave Days in Each Month

Variables	(1)	(2)	(3)	(4)	(5)	(6)
(1) Score	1.000					
(2) May Heatwaves	—	—				
(3) June Heatwaves	—	—	—			
(4) July Heatwaves	-0.0104 (0.7340)	—	—	1.000		
(5) August Heatwaves	0.0420 (0.1675)	—	—	-0.1204* (0.0000)	1.000	
(6) September Heatwaves	—	—	—	—	—	—

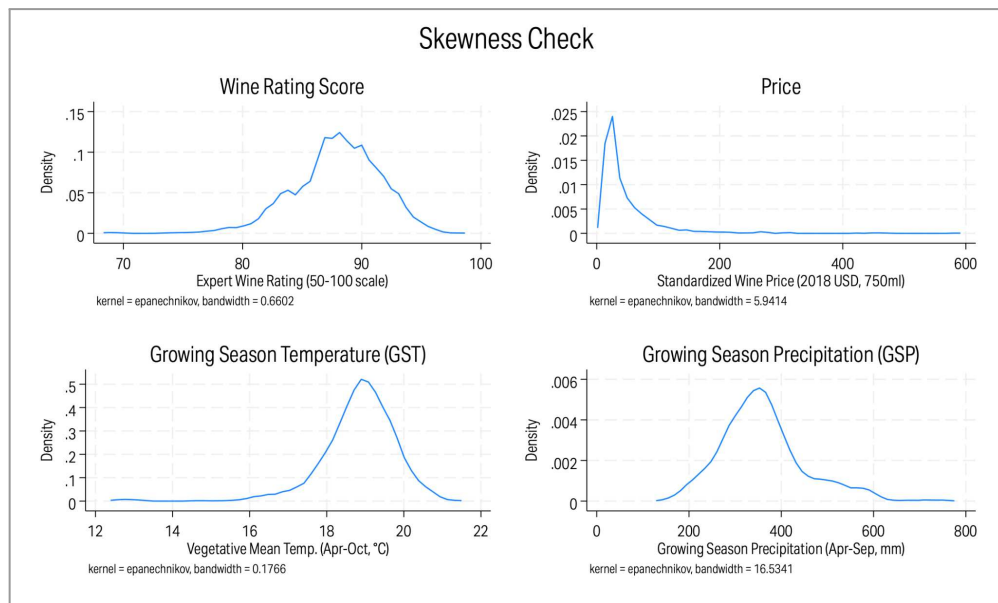
Notes: “—” means there were zero heatwaves in that month. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4.4 reveals the distribution of the monthly heatwave data points in our dataset, demonstrating that heatwaves occurred infrequently throughout the growing season. They are rare events that usually concentrate in summer (July and August) in the study period (2004-2018).

4.2.3 Skewness Check and Temporal Trends

As Figure 4.2 shows, wine prices exhibit a heavy right-hand tail, where a few prestigious labels command significant price premiums while the vast majority cluster at the lower end. This distribution of wine price exhibits positive skewness that compresses variance, and can invalidate the standard assumptions of linear models. Consequently, the natural logarithm was applied to generate the variable $\ln_price$: $\ln(P)$ to address this issue, with the aim to achieve a more symmetric distribution that stabilises the error terms. It simplifies the percentage interpretations.

Figure 4.2: Skewness check of the major variables



4.3 Econometric Models

After examining the preliminary correlation matrix in Section 4.2, several key relationships emerge. Notably, the moderate correlation between quality and price reflects the hedonic pricing framework pioneered by Oczkowski (1994) and the high correlation between thermal indices suggests that simple regressions may be subject to omitted variable bias. To address this, we employ a Fixed Effects Panel Data framework and various clustering levels to control for unobserved time-invariant characteristics, thus ensuring the robustness of our findings.

Building on the climate–viticulture models of Ashenfelter (2008), we specify two primary response functions: one for expert-rated quality and another for launch price. Models estimating quality (represented by expert rating scores) are referred to as Quality Models, while models estimating price (represented by the natural logarithm of price) are referred to as Price Models.

Our modelling strategy follows a two-stage analytical process, moving from baseline effects to complex, multi-variable interactions.

4.3.1 Stage 1: Baseline Models

In the first stage of the analysis, we adopt a strategy of testing a single heat stressor X along with one water stressor GSP to form the baseline models. The core objective is to understand how each individual climate stress indicator affects the wine's score and price. The baseline specification is as follows:

$$Y_{ijc,t} = \alpha + f(X_{c,t}) + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.1)$$

Where:

- $Y_{ijc,t}$ represents the dependent variable: either the expert rating **Score** _{ijc,t} or the natural logarithm of the release price **ln(Price)** _{ijc,t} .
- $f(X_{c,t})$ is the functional form of a **single heat stressor** being tested in comune c at time t . This represents one of the following iterations:

$$f(X_{c,t}) = \beta_1 GST_{c,t} + \beta_2 GST_{c,t}^2 \quad (4.2)$$

$$f(X_{c,t}) = \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 \quad (4.3)$$

$$f(X_{c,t}) = \beta_1 HW_{m,c,t} \quad (4.4)$$

$$f(X_{c,t}) = \beta_1 HD_{k,c,t} \quad (4.5)$$

- $GST_{c,t}$ is the average growing season temperature from April to October.
- HI is the Huglin Index.
- $HW_{m,c,t}$ is the Heatwave days in month m (May, June, July, August, September, and seasonal total from May to September, at 99th-percentile and 97.5th-percentile).
- $HD_{k,c,t}$ is the Hot Days above threshold k (32°C, 33°C, 34°C, 35°C).
- $GSP_{c,t}$ is Growing Season Precipitation (April to September).
- $Vintage_t$ is a linear time trend capturing secular progress in winemaking technology and market inflation.
- α represents the constant term.

- δ_s is the group-specific Fixed Effect, capturing unobserved time-invariant heterogeneity at the wine (i), winery (j), or comune (c) level.
- $\epsilon_{ijc,t}$ is the error term, with standard errors clustered at the level matching δ_s to account for spatial and serial correlation.
- The subscript i represents an individual wine label; j represents a winery; c represents the comune/municipality where the vineyard is located; t represents time (year); $s \in \{i, j, c\}$ and represents the level of the Fixed Effect depending on the specification (Table 4.5).

Table 4.5: Fixed Effects and Clustering Levels

Specification	Fixed Effect (δ_s)	Clustering Level	Rationale
1	Wine Label ($s=i$)	Wine Label	Controls for all time-invariant wine-specific characteristics.
2	Wine Label ($s=i$)	Winery	Controls for residuals correlated across wines from the same producer.
3	Winery ($s=j$)	Winery	Controls for winery-level management consistency and brand reputation.
4	Comune ($s=c$)	Comune	Controls for comune-level terroir and shared weather patterns.
5	Comune ($s=c$)	Winery	Controls for terroir while allowing for winery-specific error correlation.

Note: To ensure the validity of our estimates, each stressor combination is evaluated across 5 increasingly rigorous specifications. These specifications vary both the fixed effects (δ_s) and the level at which the error terms $\epsilon_{ijc,t}$ are allowed to be correlated (i.e., the clustering level).

4.3.2 Stage 2: Multi-Dimensional Comprehensive Models

In the second stage of the analysis, we combine a cumulative heat stressor C with one or two acute heat metrics A into our preferred comprehensive models. This allows us to determine if extreme heat shocks have an effect beyond the gradual thermal conditions. The specification for Stage 2 is as follows:

$$Y_{ijc,t} = \alpha + f(C_{c,t}) + f(A_{c,t}) + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6)$$

Where:

- $Y_{ijc,t}$ represents the dependent variable: either the expert rating **Score** $_{ijc,t}$ or the natural logarithm of the release price **ln(Price)** $_{ijc,t}$.
- $f(C_{c,t})$ is the functional form of a **Cumulative Heat Stressor** (either GST or HI) being tested in comune c at time t . This represents one of the following iterations:

$$f(C_{c,t}) = \beta_1 GST_{c,t} + \beta_2 GST_{c,t}^2 \quad (4.7)$$

$$f(C_{c,t}) = \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 \quad (4.8)$$

- $f(A_{c,t})$ is the functional form of the **Acute Heat Stressors** (Heatwaves or Hot Days or both) being tested in comune c at time t . This represents one of the following iterations:

$$f(A_{c,t}) = \gamma_1 HD_{k,c,t} \quad (4.9)$$

$$f(A_{c,t}) = \gamma_2 HW_{m,c,t} \quad (4.10)$$

$$f(A_{c,t}) = \gamma_1 HD_{k,c,t} + \gamma_2 HW_{m,c,t} \quad (4.11)$$

- All other variables and subscripts remain as defined in Stage 1.

Chapter 5: Results and Discussions

5.1 Climatological Variable Selection

Before evaluating the primary research questions, it is necessary to justify the selection of the explanatory bioclimatic variables utilised in the final analysis. Our selection of specific thermal thresholds and temporal windows is grounded in viticultural biology and market economics.

5.1.1 Huglin Index as Bioclimatic Baseline

Both Growing Season Temperature (GST) and the Huglin Heliothermic Index (HI) are widely established metrics used to capture the foundational heat accumulation required for grapevine development. However, this study adopts the Huglin Index as the primary cumulative thermal control for the comprehensive models due to its specific design as a bioclimatic indicator.

While GST provides a straightforward average of daily temperatures across the vegetative cycle, it treats all heat accumulation equally. In contrast, the Huglin Index is a specialised heliothermic metric designed to measure the specific thermal potential of a wine-growing region. HI explicitly calculates heat accumulation above a biologically active baseline of 10°C ([Formula 3.2](#)), effectively functioning as a targeted degree-day calculation that captures the specific thermal energy available for optimal fruit maturity. Furthermore, unlike simple temperature averages, the Huglin Index incorporates a latitude-based correction factor to account for daylight length, making it a much more precise control for assessing a vintage's baseline ripening potential.

The preference for HI over simple averages aligns with recent methodological advancements in climate-wine econometrics, such as the degree-day framework advocated by Whitnall & Alston (2025). Their research emphasises that relying exclusively on continuous seasonal averages can obscure the true physiological realities of viticulture, as simple averages fail to distinguish between the beneficial effects of moderate warmth and the detrimental effects of extreme heat. The structure of the Huglin Index mirrors this logic by isolating the beneficial baseline heat.

By establishing HI as a rigorous and precise bioclimatic baseline, the empirical models are theoretically better equipped to separate standard agricultural ripening from the acute physiological stress induced by isolated extreme weather events.

5.1.2 Physiological Resilience and Heat Thresholds

The selection of specific temperature thresholds in this study—both relative percentiles for heatwaves and absolute thresholds for isolated hot days—is grounded in the physiological resilience of grapevines and established climatological frameworks.

i. 99th-percentile vs. 97.5th-percentile

Because vineyards in Tuscany are historically adapted to the warm and dry summers typical of the Mediterranean climate (Van Leeuwen et al., 2024), they possess a baseline resilience to standard

seasonal heat. Consequently, moderate thermal extremes, such as those captured by a 97.5th-percentile threshold, frequently fall within the natural physiological tolerance of the vines. To accurately capture the effects of acute thermal shocks that overwhelm these natural defences and cause physiological damage, a more severe metric is required. Therefore, the 99th-percentile heatwave threshold is theoretically prioritised to isolate true extreme events that disrupt grapevine functioning.

ii. 33°C Threshold for Quality and 35°C Threshold for Price

The selection of distinct hot day thresholds is guided by the degree-day intervals established by Whitnall & Alston (2025), who structurally differentiate between “hot days” (30°C–35°C) that vines can temporarily withstand early in the season, and “extremely hot days” (>35°C) that trigger severe, acute physiological damage both before and after véraison. We structured our threshold selection to align with this established framework. We selected a moderate threshold of 33°C for Quality because it captures a level of heat exposure that avoids the acute thermal shock of extremely hot days (>35°C) or heatwaves. For our Price models, we selected the 35°C threshold based on a finding from Whitnall and Alston (2025) that while wine prices decline from late-season exposure to 30°C–35°C temperatures, market prices respond most severely to the extreme >35°C threshold.

5.1.3 Heatwaves in August vs. in Other Months

The temporal distribution of these thermal shocks naturally constrains the analytical focus of the models. Based on the 2004–2018 sample, the most severe thermal anomalies (99th-percentile heatwaves) are climatologically rare and exclusively concentrated in peak summer (July and August). Consequently, the months of May, June, and September experienced zero 99th-percentile heatwave events in our dataset, resulting in a natural lack of empirical variance to reliably estimate their impacts.

Early and mid-summer heatwaves did occur with sufficient frequency to be estimated, specifically 97.5th-percentile events in June, as well as both thresholds in July. However, they consistently yielded statistically insignificant coefficients in our testing models. This lack of statistical significance is consistent with viticultural findings that grapevines exhibit a comparatively lower sensitivity to acute thermal stress during their vegetative growth phase (early-to-mid summer) than they do during the late-summer ripening phase (Greer & Weedon, 2013; Poni et al., 2018; Ammoniaci et al., 2025).

Therefore, when configuring the heatwave variable for the Quality models, we primarily focus on the August véraison window.

5.1.4 Vulnerable Phenological Windows vs. Cumulative Market Risk

The selection of distinct temporal windows for assessing heatwave exposure—specifically prioritising August for Quality and the total growing season for Price—reflects the fundamental difference between the physiological sensitivity of the grapevine and the risk assessment behaviour of global wine markets.

i. August Heatwaves in Quality Models

From a viticultural perspective, the final sensory profile of a wine is highly dependent on the physiological conditions during véraison, the onset of grape ripening, which in Tuscany predominantly occurs in August. Heatwaves during this highly vulnerable developmental window induce acute

biological stress, causing direct physical damage and significant chemical degradation, such as severe reductions in acidity and the loss of essential pigment and aroma compounds (Faralli et al., 2024). Because expert evaluators judge the sensory characteristics dictated by these biological realities, the Quality model theoretically necessitates a targeted focus on August heatwaves.

ii. Total Season Heatwaves in Price Models

Conversely, market pricing mechanisms evaluate climate risk through a broader, more holistic economic lens. While physiological degradation is tied to specific phenological windows, we hypothesise that buyers, traders, and the global market evaluate the overarching, cumulative risk profile of an entire vintage. A growing season characterised by multiple extreme thermal events from May through September signals widespread environmental stress and generally difficult viticultural conditions. Therefore, to accurately capture how markets price in broad climate-related risks, the Price model utilises total-season heatwaves, reflecting the comprehensive environmental penalty applied to the vintage as a whole.

Next, we discuss three key insights from the baseline models.

5.2 Insights from Baseline Models

The Stage 1 tests—which isolate a single heat stressor (either a cumulative index, hot days, or heatwaves) alongside a water stressor (GSP)—serve as a crucial screening mechanism. Evaluating these standalone models demonstrates the limitations of a single-variable approach, and suggests that modelling weather impacts requires a multi-variable approach.

5.2.1 Cumulative Heat is the Strongest Standalone Predictor

Through the baseline models, we find that the cumulative heat index (HI or GST) is the strongest weather variable to explain wine market variations. When modelled alone, the Huglin Index yields a highly significant "inverted-U" shape (positive linear term, negative squared term) and explains 8.75% (Table 5.1, last column) of the within-group variance for Price. Similarly, for Score, the standalone Huglin Index captures 6.99% of the within-group variance (Table 5.2 on the next page, third column bottom). These cumulative models substantially outperform standalone models of extreme, acute shocks.

The baseline heatwave tests yielded statistically insignificant results. None of the 97.5th-percentile heatwaves yielded significant results. Isolated extreme heat variables (such as August heatwave or hot days above 35°C) exhibit low explanatory power when modelled independently.

Stage 1 heatwave-only and hot-days-only tests demonstrate that forcing a single heat variable to explain the entire vintage effect leads to reduced explanatory power. Because a standalone heat variable is forced to capture *both* the positive ripening effects of a generally warm year and the negative damage of the specific extreme event, its coefficient captures confounding effects. Consequently, the extreme heat variables (Hot Days > 35°C and August 99th-percentile Heatwaves) become statistically insignificant under the wine-level fixed effects (Table 5.2 on the next page, last two columns).

Table 5.1: GST and HI in Baseline Models for Price as the Single Cumulative Heat Stressor

	GST-only Model	HI-only Model
GST	0.83963** (0.38284)	
GST^2	-0.02161** (0.01011)	
HI		0.00216** (0.00102)
HI^2		-0.00000* (0.00000)
GSP	-0.00027 (0.00054)	-0.00023 (0.00062)
GSP^2	0.00000 (0.00000)	0.00000 (0.00000)
R-squared	0.97217	0.97175
R-squared (within)	0.08578	0.08754
Observations	461	453

Both models employ a specification of wine fixed effects with wine cluster. Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 5.2: Baseline Models for Score: Testing a Single Heat Stressor and One Water Stressor

	GST-only Model	HI-only Model	Hot Days > 33°C only	Hot Days > 35°C only	Heatwaves-only (Aug, 99-pctl)
GST	13.35410** (5.85925)				
GST^2	-0.32288** (0.15356)				
HI		0.03781** (0.01859)			
HI^2		-0.00001* (0.00000)			
Hot Days > 33°C			0.05931** (0.02532)		
Hot Days > 35°C				0.05686 (0.07202)	
Aug Heatwaves (99-pctl)					-0.06241 (0.29986)
GSP	-0.00728 (0.00820)	-0.00295 (0.00924)	-0.00062 (0.00846)	-0.00293 (0.00871)	-0.00506 (0.00848)
GSP^2	0.00001 (0.00001)	0.00001 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)
R-squared (within)	0.05438	0.06997	0.04609	0.02678	0.02406
Observations	499	491	499	499	499

Notes: All models in this table employ a wine-level fixed effects specification with standard errors clustered at the winery level. Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

5.2.2 The Role of Precipitation is Region-Dependent, Not Wine-Dependent

Across all Stage 1 standalone models, Growing Season Precipitation (GSP) behaves consistently: it is insignificant in the strict Wine FE and Winery FE specifications. However, GSP frequently becomes positive and significant in the broader Comune FE specifications. This suggests that rainfall amounts do not meaningfully explain the year-to-year price or score fluctuations for a *specific* wine brand; but at a macro level, greater precipitation is generally beneficial for a region or municipality's aggregate pricing, even if it does not drive the micro-level variance of individual top-tier wines.

5.2.3 The Limitation of Single Stressors

Collectively, Stage 1's baseline tests validate the methodological need for Stage 2. These results indicate that a simple, single-stressor model may be insufficient to adequately capture the economics of the wine market because weather impacts are a combination of necessary baseline warmth (best captured by HI or GST) and the distinct, acute damages of isolated extremes (HD and HW).

5.3 Comprehensive Models Led by Cumulative Heat Indicator

Stage 1 has established a baseline by evaluating single-stressor models. The data demonstrates that isolated extreme heat variables yield diluted significance when modelled without cumulative baseline controls. For example, when a 99th-percentile August heatwave is modelled entirely in isolation against price, it yields a statistically insignificant coefficient (-0.020) and explains only 7.77% of the within-group variance. To correct this omitted variable bias, the study developed two Comprehensive Models integrating Growing Season Temperature (GST) alongside acute daily spikes (Hot Days) and prolonged heatwaves: one for Score and one for Price.

- **The Comprehensive Quality Model (Score):** This model integrates a cumulative heat indicator (GST or HI) alongside the moderate heat indicator Hot Days above 33°C and 99th-percentile August heatwaves.

- **The Comprehensive Market Model (Price):** This model integrates a cumulative heat indicator (GST or HI) alongside the extreme heat threshold Hot Days above 35°C and the 99th-percentile total season heatwaves from May to September.

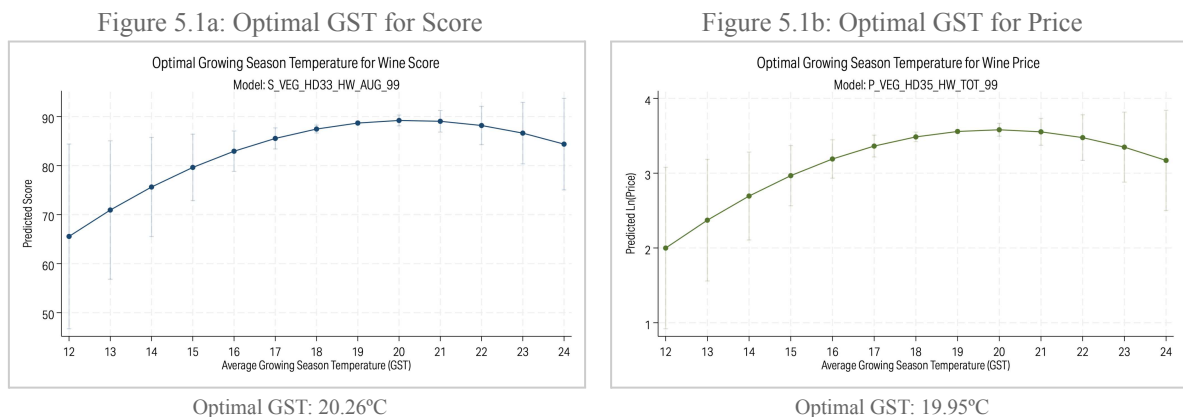
5.3.1 Optimal Growing Season Temperature for Quality and Price

When incorporated into the comprehensive Quality Model, GST exhibits the significant "inverted-U" shape again and the model explains 7.21% of the within-group variance (Table 5.3, next page, column 1-2). This suggests an estimated optimal GST for wine score exists, which is approximately 20.26°C (Figure 5.1a). This suggests that moderate heat accumulation elevates expert ratings, but excessive heat accumulation is associated with a decline in sensory wine quality.

The comprehensive Price Model integrates GST alongside extreme daily heat and total season heatwaves. It explains 10.98% of the within-group variance (Table 5.4, column 1-2, second last row). The GST linear term and the squared term are both significant (0.99780** and -0.02500**, respectively, Table 5.4, column 1-2). This model indicates an estimated optimal thermal threshold for Price at approximately 19.95°C (Figure 5.1b).

Furthermore, controlling for the negative penalty of total season heatwaves (-0.06320***) reveals a positive and significant association between price and isolated days above 35°C (0.00890**), as Table 5.4 (column 1 & 2) shows. It implies that brief heat spikes might not be detrimental provided they do not compound into prolonged heatwave events. Alternatively, this positive coefficient might reflect complex market dynamics where buyers associate certain heat spikes with prestigious “warm” vintages. This finding mirrors the one in the Comprehensive Quality Model: hot days above 33°C demonstrates a mild positive effect (Table 5.3, next page, column 1 & 2) when controlling for heatwave events.

Figure 5.1: Optimal Cumulative Heat for Wine Score and Price



Notes: a) this model computes average growing season temperature GST, Hot days > 33°C, August 99th-percentile heatwaves, and GSP. b) this model computes GST, Hot days > 35°C, and 99th-percentile total season heatwaves from May to September, and GSP.

Table 5.3: Comprehensive Quality Model Led By Growing Season Temperature (GST)

	(1) Wine FE Wine Cluster	(2) Wine FE Winery Cluster	(3) Winery FE Winery Cluster	(4) Comune FE Comune Cluster	(5) Comune FE Winery Cluster
GST	14.06117** (6.92589)	14.06117** (5.92762)	1.88360 (5.19435)	-0.98483 (4.31518)	-0.98483 (4.98391)
GST^2	-0.34702* (0.18041)	-0.34702** (0.15394)	-0.02766 (0.13682)	0.04908 (0.11377)	0.04908 (0.13093)
Hot Days > 33°C	0.05225** (0.02325)	0.05225** (0.02539)	-0.00694 (0.02427)	-0.00403 (0.02182)	-0.00403 (0.02194)
August Heatwaves (99-pctl)	-0.52832* (0.27746)	-0.52832* (0.28564)	-0.53372* (0.28184)	-0.59722** (0.23888)	-0.59722** (0.27080)
GSP	-0.00536 (0.00828)	-0.00536 (0.00817)	0.00089 (0.00642)	-0.00178 (0.00604)	-0.00178 (0.00639)
GSP^2	0.00001 (0.00001)	0.00001 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)
R-squared (within)	0.07215	0.07215	0.00867	0.03063	0.03063
Observations	499	499	917	1068	1068

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. This model computes average growing season temperature (GST), Hot days > 33°C, and August 99th-percentile heatwaves, across 5 FE specifications.

Table 5.4: Comprehensive Price Model Led By Growing Season Temperature (GST)

	(1) Wine FE Wine Cluster	(2) Wine FE Winery Cluster	(3) Winery FE Winery Cluster	(4) Comune FE Comune Cluster	(5) Comune FE Winery Cluster
GST	0.99780** (0.40551)	0.99780** (0.40792)	-1.46169* (0.83104)	-1.35346 (0.88651)	-1.35346 (0.89800)
GST^2	-0.02500** (0.01074)	-0.02500** (0.01082)	0.04008* (0.02250)	0.03886 (0.02367)	0.03886 (0.02392)
Hot Days > 35°C	0.00890** (0.00380)	0.00890** (0.00391)	0.00361 (0.00961)	-0.00330 (0.00953)	-0.00330 (0.00987)
Total Heatwaves (99-pctl)	-0.06320*** (0.02257)	-0.06320*** (0.02177)	-0.08534* (0.04692)	-0.04055 (0.04717)	-0.04055 (0.04949)
GSP	-0.00013 (0.00053)	-0.00013 (0.00054)	0.00244 (0.00149)	0.00322** (0.00135)	0.00322** (0.00142)
GSP^2	0.00000 (0.00000)	0.00000 (0.00000)	-0.00000* (0.00000)	-0.00000** (0.00000)	-0.00000** (0.00000)
R-squared (within)	0.10985	0.10985	0.05387	0.02425	0.02425
Observations	461	461	839	984	984

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. This model computes average growing season temperature (GST), Hot days > 35°C, and 99th-percentile total season heatwaves from May to September, and GSP, across 5 FE specifications.

5.3.2 The Impact of Growing Season Precipitation

In this section, we discuss the role of the water stressor as a macro-regional driver, utilising the column-by-column variations generated by the fixed-effects methodology. GSP serves as an essential environmental control in our models, and its behaviour across the different fixed-effects specifications provides valuable regional insights. Rather than driving the year-over-year variation for a specific wine, the empirical evidence indicates that precipitation functions as a macro-level regional driver, or at least at the municipality level.

Under the strictest wine-level fixed effects specifications, GSP consistently returns a statistically insignificant coefficient (Table 5.5, column 1-3). This suggests that for a specific, top-tier wine brand, annual fluctuations in total rainfall do not meaningfully indicate price or expert score variations. However, when the models are broadened to the municipality level (comune fixed effects, Table 5.5, column 4-5), the linear GSP term frequently becomes positive and statistically significant (e.g., 0.00331** in comprehensive price models).

Our findings about GSP suggest that while individual elite producers may mitigate rainfall variations through specific viticultural practices (e.g., canopy management or precise harvest timing), at an aggregate regional level, adequate precipitation remains a significant beneficial factor for overall market pricing.

Table 5.5: Comprehensive Price Model Led By Huglin Index (HI)

	(1) Wine FE Wine Cluster	(2) Wine FE Winery Cluster	(3) Winery FE Winery Cluster	(4) Comune FE Comune Cluster	(5) Comune FE Winery Cluster
HI	0.00378*** (0.00116)	0.00378*** (0.00115)	-0.00166 (0.00242)	-0.00075 (0.00245)	-0.00075 (0.00258)
HI^2	-0.00000*** (0.00000)	-0.00000*** (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
Hot Days > 35°C	0.01217*** (0.00430)	0.01217*** (0.00437)	0.00349 (0.01079)	-0.00322 (0.01035)	-0.00322 (0.01075)
Total Heatwaves (99-pctl)	-0.07329*** (0.02050)	-0.07329*** (0.01930)	-0.06874 (0.04355)	-0.02333 (0.04430)	-0.02333 (0.04633)
GSP	-0.00010 (0.00059)	-0.00010 (0.00061)	0.00227 (0.00169)	0.00331** (0.00152)	0.00331** (0.00155)
GSP^2	0.00000 (0.00000)	0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000** (0.00000)	-0.00000** (0.00000)
R-squared (within)	0.12067	0.12067	0.04736	0.01908	0.01908
Observations	453	453	827	972	972

Notes: This model computes HI, Hot days > 35°C, 99th-percentile total season heatwaves from May to September, and GSP. Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

5.3.3 The August Heatwave Penalty

The primary empirical contribution of this research is quantifying the estimated negative effects of extreme weather events during the most vulnerable physiological window: August (véraison). The models reveal a divergence between how expert critics and wine markets (or wine producers and whoever has the right to set the release price) assess this risk.

i. The Effect of August Heatwaves on Wine Quality Scores

Expert evaluators appear resilient to moderate extreme heat, systematically penalising a vintage only when the most extreme thermal shocks occur. In the Comprehensive Quality Model led by Huglin Index, the 99th-percentile August heatwave is associated with a notable decline in sensory evaluations (-0.586**), capturing 8.57% of the within-group variance (Table 5.6, column 1).

Similarly, the Comprehensive Quality Model led by GST also demonstrates that critics' scores are negatively affected by the 99th-percentile August heatwaves (-0.52832*, Table 5.3, column 1). The

increase in coefficient magnitude, as well as in level of significance (from $|-0.52832^*|$ to $|-0.59722^{**}|$, Table 5.3, middle row), suggests that the negative impact of August heatwaves is more pronounced at the aggregate municipality/comune level than for an individual wine. While a top-tier wine might maintain higher scores during a heatwave due to its established prestige and resources, the municipality as a whole exhibits a more significant decline in aggregate ratings. This result reveals a regional geographic vulnerability to the August heatwave, as the evidence indicates it is a substantial environmental stressor associated with a negative shift in the overall distribution of ratings for a given vintage and location.

Our finding about August heatwaves aligns with viticultural biology. August coincides with *véraison*, the onset of ripening. The data provides empirical evidence that heatwaves during this specific window are associated with a noticeable reduction in sensory evaluations, likely due to physiological degradation.

Table 5.6: Comprehensive Quality Model Led By Huglin Index (HI)

	(1) Wine FE Wine Cluster	(2) Wine FE Winery Cluster	(3) Winery FE Winery Cluster	(4) Comune FE Comune Cluster	(5) Comune FE Winery Cluster
HI	0.04326** (0.02127)	0.04326** (0.01835)	0.00569 (0.01519)	0.00070 (0.01118)	0.00070 (0.01415)
HI ²	-0.00001* (0.00000)	-0.00001** (0.00000)	-0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
Hot Days > 33°C	0.04497* (0.02464)	0.04497* (0.02453)	-0.00930 (0.02498)	-0.00638 (0.02327)	-0.00638 (0.02267)
Aug Heatwaves (99-pctl)	-0.58608** (0.26853)	-0.58608** (0.27860)	-0.51638* (0.28019)	-0.58060** (0.23541)	-0.58060** (0.26712)
GSP	-0.00320 (0.00897)	-0.00320 (0.00913)	0.00345 (0.00688)	0.00099 (0.00617)	0.00099 (0.00654)
GSP ²	0.00001 (0.00001)	0.00001 (0.00001)	-0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)
R-squared (within)	0.08574	0.08574	0.00843	0.03093	0.03093
Observations	491	491	905	1056	1056

Notes: This model computes Huglin Index (HI), Hot days > 33°C, August 99th-percentile heatwaves, and GSP. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

ii. The Impact of Total Season Heatwaves on Wine Prices

For buyers and traders, the total cumulative exposure to extreme heatwaves over the season is a significant factor associated with price reductions. The comprehensive Price Model led by HI achieves the maximum explanatory power (12.06% Within R-squared) for 99th-percentile total season heatwaves, with a highly negative coefficient (-0.073^{***} , Table 5.5, column 1). For comparison, the 99th-percentile August heatwave yields a less substantial coefficient (-0.049^{**}) while explaining only 10.23% of the within-group price variance when tested alongside HI, Hot Days > 35°C, and GSP.

These results suggest that markets evaluate broad vintage risk (total season), whereas sensory evaluations are more likely to be tethered to the physiological constraints of *véraison* (August).

5.4 Inland vs. Coastal Macro-Terroirs

The comparison between coastal and inland macro-terroirs suggests a clear geographical differentiation in the response to thermal stress, consistent with the presence of coastal moderation effects and stronger heat accumulation in inland areas.

Applying the preferred fixed-effects specifications (comune fixed effects for Quality and winery fixed effects for Price), the results can be summarised as follows:

5.4.1 Quality Effects: Agronomic Divergence

Using comune-level fixed effects, the estimates indicate that the sensory quality of inland and coastal vineyards responds differently to late-season heat stress.

- i. Inland areas:* Exposure to 99th-percentile August heatwaves is associated with a statistically significant decline in expert ratings for inland wines (coefficient -0.64207^{**} , Table 5.7, left panel, middle row highlighted in bold), suggesting greater vulnerability to extreme heat during critical phenological stages.
- ii. Coastal areas:* For coastal wines, the estimated effect of the same heatwave variable is smaller and statistically insignificant (coefficient -0.2021 , Table 5.7, left panel, middle row highlighted in bold), consistent with a moderating influence from maritime conditions.

Table 5.7: Macro-Terroirs' Responses to Heatwaves

Quality Model (Comune FE)			Price Model (Winery FE)		
	Inland	Coastal		Inland	Coastal
HI	0.00568 (0.01494)	-0.03277 (0.02342)	HI	-0.00343 (0.00300)	0.00780 (0.00718)
HI ²	-0.00000 (0.00000)	0.00001 (0.00001)	HI ²	0.00000 (0.00000)	-0.00000 (0.00000)
Hot Days > 33°C	-0.00607 (0.03037)	-0.02772 (0.04373)	Hot Days > 35°C	0.00774 (0.01363)	0.00867 (0.02347)
Aug Heatwaves (99-pctl)	-0.64207** (0.31084)	-0.20210 (0.40130)	Total Heatwaves (99-pctl)	-0.14492** (0.07274)	0.00037 (0.05702)
GSP	-0.00140 (0.01098)	0.01943 (0.01219)	GSP	0.00365 (0.00220)	-0.00094 (0.00244)
GSP ²	0.00000 (0.00001)	-0.00003 (0.00002)	GSP ²	-0.00001** (0.00000)	0.00000 (0.00000)
R-squared (within)	0.03424	0.02714	R-squared (within)	0.06918	0.05146
Observations	604	369	Observations	465	297

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Quality Model computes Huglin Index (HI), Hot days > 33°C, August 99th-percentile heatwaves, and GSP; Comune-level fixed effects; Standard errors clustered at the comune level. Price Model computes HI, Hot days > 35°C, 99th-percentile total season heatwaves from May to September, and GSP; Winery-level fixed effects; Standard errors clustered at the winery level.

5.4.2 Price Effects: Market Differentiation

The market response, estimated using winery fixed effects, exhibits a differentiated pattern across the two macro-terroirs.

- i. Inland wines:* Exposure to 99th-percentile total-season heatwaves is associated with a substantial and statistically significant reduction in prices (coefficient -0.14492^{**} , Table 5.7, right panel, middle row highlighted in bold).

- ii.* **Coastal wines:** For coastal wines, the estimated effect is close to zero and statistically insignificant (coefficient 0.00037, Table 5.7, right panel, middle row highlighted in bold), indicating no detectable price response to the same thermal stress.

Taken together with the previous results, these findings suggest that geographical location plays a pronounced role. In particular, coastal areas exhibit both weaker quality impacts and limited price adjustments, whereas inland areas experience both stronger quality deterioration and larger price declines.

Overall, the coastal–inland comparison highlights the importance of spatial climatic variation in shaping both production outcomes and market valuation, suggesting that market responses are aligned with underlying regional exposure to thermal stress.

5.5 PDO vs. PGI Designations

The comparison between Protected Designation of Origin (PDO) and Protected Geographical Indication (PGI) wines suggests a divergence between the role of regulatory stringency in maintaining wine quality and the way markets price climate-related risks.

5.5.1 PDO Regulations and Wine Quality

PDO regulations (which encompass Italy's DOC and DOCG classifications) impose stricter production constraints, including limited or prohibited irrigation, whereas PGI (IGT) regulations allow greater flexibility. Despite these restrictions, the results indicate that PDO frameworks are associated with greater resilience in maintaining measured wine quality during acute thermal stress.

According to the Quality Models in Table 5.8 (left panel, middle row, highlighted in bold), exposure to 99th-percentile August heatwaves is associated with a large and statistically significant decline in expert ratings for PGI wines (coefficient -1.06618^{***}). By contrast, the estimated effect for PDO wines is small and statistically insignificant (coefficient -0.03498).

Table 5.8: GI Wines' Responses to Heatwaves

Quality Model			Price Model		
	PDO	PGI		PDO	PGI
HI	0.02148 (0.03011)	0.08241*** (0.02965)	HI	0.00201 (0.00142)	0.00532*** (0.00162)
HI^2	-0.00000 (0.00001)	-0.00002** (0.00001)	HI^2	-0.00000 (0.00000)	-0.00000*** (0.00000)
Hot Days > 33°C	-0.00235 (0.03211)	0.09949*** (0.03674)	Hot Days > 35°C	0.01023* (0.00598)	0.01429** (0.00566)
Aug Heatwaves (99-pctl)	-0.03498 (0.42427)	-1.06618*** (0.32206)	Total Heatwaves (99-pctl)	-0.06728*** (0.02558)	-0.07063** (0.03023)
GSP	-0.01197 (0.01104)	0.00906 (0.01605)	GSP	-0.00079 (0.00055)	0.00126 (0.00111)
GSP^2	0.00002 (0.00001)	-0.00000 (0.00002)	GSP^2	0.00000 (0.00000)	-0.00000 (0.00000)
R-squared (within)	0.10925	0.13597	R-squared (within)	0.23903	0.11775
Observations	267	224	Observations	245	208

Notes: All models here employ wine-level FE with wine-level clustering. Quality Model computes Huglin Index (HI), Hot days > 33°C, August 99th-percentile heatwaves, and GSP. Price Model computes HI, Hot days > 35°C, 99th-percentile total season heatwaves from May to September, and GSP. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.5.2 Reputational Price Protection

While the PDO framework appears to protect the physical quality of the wine, the results do not support the hypothesis that a prestigious PDO label provides a reputational price protection effect in the market. The price models indicate that extreme heat is associated with significant price reductions regardless of official designation. Exposure to total-season 99th-percentile heatwaves leads to substantial and statistically significant price declines in both categories: approximately 6.7% for PDO wines (coefficient -0.06728^{***}) and around 7.0% for PGI wines (coefficient -0.07063^{**}) as Table 5.8 shows (right panel, middle row highlighted in bold). These findings suggest that markets

systematically discount wine prices following severe thermal events, even when measured sensory quality remains unaffected. In this sense, prices appear to reflect broader perceptions of vintage-level risk rather than fully incorporating the mitigating role of PDO production standards. This dynamic highlights a potential economic vulnerability within traditional appellation systems in the context of a changing climate.

Chapter 6: Conclusion

This thesis provides an empirical analysis of how climate change, characterised by rising baseline temperatures and an increased frequency of extreme thermal events, affects the Tuscan wine sector. By integrating agronomic climate indicators with hedonic pricing methodologies, the study examines the mechanisms through which weather variability influences both perceived wine quality and market valuation.

The results indicate that moderate heat accumulation is associated with improvements in both wine quality and price up to an estimated optimal threshold (approximately 19.95°C to 20.26°C), beyond which additional heat exposure is associated with declining outcomes. In particular, extreme thermal events are found to have negative effects on both quality and price.

The analysis also points to a divergence in how expert evaluations and market outcomes reflect climate risk. Expert ratings respond most strongly to late-season heatwaves, particularly those occurring during the *véraison* period. By contrast, market prices appear to reflect cumulative exposure to heat stress over the full growing season, suggesting that pricing incorporates broader vintage-level risk considerations.

In addition, the results highlight spatial and regulatory heterogeneity within Tuscany. Coastal vineyards exhibit weaker responses to thermal stress in both quality and price models, consistent with a moderating maritime influence. Inland areas, by contrast, show stronger negative effects, indicating greater exposure to heat-related risks. With respect to GI frameworks, PDO wines are associated with more stable quality outcomes relative to PGI wines under extreme heat conditions. A corresponding mitigation is not observed in price responses, as prices for both PDO and PGI designations show a comparable decrease.

While the study provides a consistent empirical framework, it is subject to several limitations, including the 2004–2018 observation period and the use of aggregated seasonal precipitation measures. Future research could extend this analysis by incorporating higher-frequency hydrological data to better capture the timing and intensity of precipitation shocks. Expanding the temporal scope to include more recent vintages (e.g., 2019–2021) would also be valuable, particularly as updated land cover datasets (e.g., CLC 2024) become available.

Overall, the findings suggest that the economic implications of climate change for Tuscan wine production depend not only on biophysical impacts but also on how these risks are perceived and incorporated into market valuations.

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Appendix A: Baseline Models & Screening Results

A.1 Standalone Huglin Index

$$Score_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.1a)$$

$$\ln(Price)_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.1b)$$

Table A.1: Baseline Models – Standalone Huglin Index (HI)

	Quality Models					Price Models				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
HI	0.03781* (0.02164)	0.03781** (0.01859)	0.00508 (0.01505)	-0.00005 (0.01097)	-0.00005 (0.01401)	0.00216** (0.00102)	0.00216* (0.00110)	-0.00200 (0.00219)	-0.00051 (0.00231)	-0.00051 (0.00239)
HI^2	-0.00001 (0.00000)	-0.00001* (0.00000)	-0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	-0.00000* (0.00000)	-0.00000* (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
GSP	-0.00295 (0.00906)	-0.00295 (0.00924)	0.00449 (0.00690)	0.00242 (0.00623)	0.00242 (0.00654)	-0.00023 (0.00062)	-0.00023 (0.00063)	0.00224 (0.00169)	0.00337** (0.00154)	0.00337** (0.00156)
GSP^2	0.00001 (0.00001)	0.00001 (0.00001)	-0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00000)	0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000** (0.00000)	-0.00000** (0.00000)
R-squared (within)	0.06997	0.06997	0.00356	0.02599	0.02599	0.08754	0.08754	0.04472	0.01856	0.01856
Observations	491	491	905	1056	1056	453	453	827	972	972

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Panel fixed effects specifications: (1) wine FE with wine clustering; (2) wine FE with winery clustering; (3) winery FE with winery clustering; (4) comune FE with comune clustering; (5) comune FE with winery clustering. GSP is growing season precipitation.

A.2 Standalone Growing Season Temperature (GST)

$$Score_{ijc,t} = \alpha + \beta_1 GST_{c,t} + \beta_2 GST_{c,t}^2 + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.1c)$$

$$\ln(Price)_{ijc,t} = \alpha + \beta_1 GST_{c,t} + \beta_2 GST_{c,t}^2 + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.1d)$$

Table A.2: Baseline Models – Standalone Growing Season Temperature (GST)

	Quality Models					Price Models				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
GST	13.35410* (6.98987)	13.35410** (5.85925)	0.89176 (5.12789)	-2.27568 (4.23911)	-2.27568 (4.87026)	0.83963** (0.38284)	0.83963** (0.39035)	-1.45846* (0.78212)	-1.28085 (0.85925)	-1.28085 (0.85776)
GST^2	-0.32288* (0.18308)	-0.32288** (0.15356)	-0.00973 (0.13526)	0.07441 (0.11148)	0.07441 (0.12806)	-0.02161** (0.01011)	-0.02161** (0.01030)	0.03832* (0.02090)	0.03588 (0.02286)	0.03588 (0.02266)
GSP	-0.00728 (0.00824)	-0.00728 (0.00820)	0.00350 (0.00635)	0.00122 (0.00610)	0.00122 (0.00631)	-0.00027 (0.00054)	-0.00027 (0.00054)	0.00263* (0.00148)	0.00348** (0.00138)	0.00348** (0.00142)
GSP^2	0.00001 (0.00001)	0.00001 (0.00001)	-0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00000)	0.00000 (0.00000)	-0.00000* (0.00000)	-0.00000** (0.00000)	-0.00000** (0.00000)
R-squared (within)	0.05438	0.05438	0.00376	0.02571	0.02571	0.08578	0.08578	0.05007	0.02313	0.02313
Observations	499	499	917	1068	1068	461	461	839	984	984

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Panel fixed effects specifications: (1) wine FE with wine clustering; (2) wine FE with winery clustering; (3) winery FE with winery clustering; (4) comune FE with comune clustering; (5) comune FE with winery clustering. GSP is growing season precipitation.

A.3 Standalone Hot Days Thresholds

$$Score_{ijc,t} = \alpha + \gamma_1 HD_{32,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.1e)$$

$$\ln(Price)_{ijc,t} = \alpha + \gamma_1 HD_{35,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.1f)$$

Table A.3: Baseline Models – Standalone Hot Days (33°C for Quality, 35°C for Price)

	Quality Models					Price Models				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
Hot Days	0.05931*** (0.02224)	0.05931** (0.02532)	-0.00139 (0.02237)	0.00313 (0.01907)	0.00313 (0.01966)	0.00298 (0.00385)	0.00298 (0.00395)	-0.00045 (0.00893)	-0.00033 (0.00877)	-0.00033 (0.00868)
	> 33°C	> 33°C	> 33°C	> 33°C	> 33°C	> 35°C	> 35°C	> 35°C	> 35°C	> 35°C
GSP	-0.00062 (0.00839)	-0.00062 (0.00846)	0.00335 (0.00654)	0.00086 (0.00606)	0.00086 (0.00634)	-0.00000 (0.00057)	-0.00000 (0.00060)	0.00236 (0.00149)	0.00319** (0.00138)	0.00319** (0.00143)
GSP^2	0.00000 (0.00001)	0.00000 (0.00001)	-0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00000)	0.00000 (0.00000)	-0.00000* (0.00000)	-0.00000** (0.00000)	-0.00000** (0.00000)
R-squared (within)	0.04609	0.04609	0.00129	0.02302	0.02302	0.07666	0.07666	0.04708	0.01936	0.01936
Observations	499	499	917	1068	1068	461	461	839	984	984

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Panel fixed effects specifications: (1) wine FE with wine clustering; (2) wine FE with winery clustering; (3) winery FE with winery clustering; (4) comune FE with comune clustering; (5) comune FE with winery clustering. GSP is growing season precipitation.

A.4 Standalone Heatwaves

$$Score_{ijc,t} = \alpha + \beta_1 HW_{aug,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.1g)$$

$$\ln(Price)_{ijc,t} = \alpha + \beta_1 HW_{tot,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.1h)$$

Table A.4: Baseline Models – Standalone Heatwaves (August and Seasonal Total)

	Quality Models					Price Models				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
Heatwaves	-0.06241 (0.27579)	-0.06241 (0.29986)	-0.36351 (0.28016)	-0.40343* (0.22699)	-0.40343 (0.26118)	-0.03356* (0.01914)	-0.03356* (0.01807)	-0.05388 (0.03924)	-0.00466 (0.03845)	-0.00466 (0.04026)
	August	August	August	August	August	Total	Total	Total	Total	Total
GSP	-0.00506 (0.00842)	-0.00506 (0.00848)	0.00204 (0.00635)	-0.00112 (0.00608)	-0.00112 (0.00632)	-0.00018 (0.00055)	-0.00018 (0.00055)	0.00215 (0.00145)	0.00318** (0.00135)	0.00318** (0.00140)
GSP^2	0.00000 (0.00001)	0.00000 (0.00001)	-0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00000)	0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000** (0.00000)	-0.00000** (0.00000)
R-squared (within)	0.02406	0.02406	0.00360	0.02545	0.02545	0.08400	0.08400	0.04939	0.01937	0.01937
Observations	499	499	917	1068	1068	461	461	839	984	984

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Panel fixed effects specifications: (1) wine FE with wine clustering; (2) wine FE with winery clustering; (3) winery FE with winery clustering; (4) comune FE with comune clustering; (5) comune FE with winery clustering. GSP is growing season precipitation. “Total” in the Heatwaves row denotes the 99th-percentile total season heatwaves from May to September.

Appendix B: Robustness Checks

B.1 Robustness Checks – Hot Day Thresholds

$$Score_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \gamma_1 HD_{k,c,t} + \gamma_2 HW_{aug,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6a)$$

$$\ln(Price)_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \gamma_1 HD_{k,c,t} + \gamma_2 HW_{tot,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6b)$$

Table B.1: Robustness Checks – Hot Day Thresholds

	Quality Models				Price Models			
Threshold Tested:	HD > 32°C	HD > 33°C	HD > 34°C	HD > 35°C	HD > 32°C	HD > 33°C	HD > 34°C	HD > 35°C
Hot Days	0.03224 (0.01952)	0.04497* (0.02464)	0.06608* (0.03811)	0.09071 (0.07087)	-0.00045 (0.00157)	0.00097 (0.00203)	0.00472 (0.00346)	0.01217*** (0.00430)
Heatwaves (99th-pctl)	-0.57723** (0.26472)	-0.58608** (0.26853)	-0.64133** (0.27560)	-0.65770** (0.29529)	-0.05070** (0.01966)	-0.05661*** (0.01991)	-0.06651*** (0.02104)	-0.07329*** (0.02050)
	<i>August</i>	<i>August</i>	<i>August</i>	<i>August</i>	<i>Total</i>	<i>Total</i>	<i>Total</i>	<i>Total</i>
HI	0.04146* (0.02133)	0.04326** (0.02127)	0.04827** (0.02193)	0.04920** (0.02314)	0.00237** (0.00104)	0.00251** (0.00108)	0.00309** (0.00119)	0.00378*** (0.00116)
HI^2	-0.00001* (0.00000)	-0.00001* (0.00000)	-0.00001* (0.00000)	-0.00001* (0.00001)	-0.00000* (0.00000)	-0.00000** (0.00000)	-0.00000** (0.00000)	-0.00000*** (0.00000)
GSP	-0.00392 (0.00896)	-0.00320 (0.00897)	-0.00304 (0.00894)	-0.00309 (0.00891)	-0.00020 (0.00061)	-0.00019 (0.00061)	-0.00014 (0.00060)	-0.00010 (0.00059)
GSP^2	0.00001 (0.00001)	0.00001 (0.00001)	0.00001 (0.00001)	0.00001 (0.00001)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
R-squared (within)	0.08461	0.08574	0.08457	0.08150	0.10568	0.10612	0.11218	0.12067
Observations	491	491	491	491	453	453	453	453

Notes: All models in this table employ wine-level fixed effects, with standard errors clustered at the wine level. Standard errors are reported in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. HD denotes hot days. HI denotes the Huglin Index. GSP denotes growing season precipitation. “Total” in the Heatwaves row denotes the 99th-percentile total season heatwaves from May to September.

B.2 Robustness Checks – Huglin Index vs. GST

$$Score_{ijc,t} = \alpha + \beta_1 GST_{c,t} + \beta_2 GST_{c,t}^2 + \gamma_1 HD_{33,c,t} + \gamma_2 HW_{aug,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6c)$$

$$Score_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \gamma_1 HD_{33,c,t} + \gamma_2 HW_{aug,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6d)$$

$$\ln(Price)_{ijc,t} = \alpha + \beta_1 GST_{c,t} + \beta_2 GST_{c,t}^2 + \gamma_1 HD_{35,c,t} + \gamma_2 HW_{tot,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6e)$$

$$\ln(Price)_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \gamma_1 HD_{35,c,t} + \gamma_2 HW_{tot,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6f)$$

Table B.2: Comparing Model Robustness Between GST-led Models and HI-led Models

Variables	GST-led Quality Model (a)	HI-led Quality Model (b)	Variables	GST-led Price Model (c)	HI-led Price Model (d)
Hot Days > 33°C	0.05225** (0.02325)	0.04497* (0.02464)	Hot Days > 35°C	0.00890** (0.00380)	0.01217*** (0.00430)
Aug Heatwaves (99-pctl)	-0.52832* (0.27746)	-0.58608** (0.26853)	Total Heatwaves (99-pctl)	-0.06320*** (0.02257)	-0.07329*** (0.02050)
GSP	-0.00536 (0.00828)	-0.00320 (0.00897)	GSP	-0.00013 (0.00053)	-0.00010 (0.00059)
GSP^2	0.00001 (0.00001)	0.00001 (0.00001)	GSP^2	0.00000 (0.00000)	0.00000 (0.00000)
R-squared (within)	0.07215	0.08574	R-squared (within)	0.10985	0.12067
Observations	499	491	Observations	461	453

Notes: All models control for wine-level FE with wine-level clustering. (a) This model computes average growing season temperature (GST), Hot days > 33°C, August 99th-percentile heatwaves, and GSP. (b) This model computes Huglin Index (HI), Hot days > 33°C, August 99th-percentile heatwaves, and GSP. (c) This model computes GST, Hot days > 35°C, 99th-percentile total season heatwaves from May to September, and GSP. (d) This model computes HI, Hot days > 35°C, 99th-percentile total season heatwaves from May to September, and GSP. Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

B.3 Quality Model – Comparing 97.5th-percentile vs. 99th-percentile August Heatwaves

$$Score_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \gamma_1 HD_{33,c,t} + \gamma_2 HW_{aug,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6d')$$

Table B.3: Quality Model – Comparing 97.5th-percentile vs. 99th-percentile August Heatwaves

	Quality Model (97.5th-percentile August Heatwaves)					Quality Model (99th-percentile August Heatwaves)				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
HI	0.04157*	0.04157**	0.00386	-0.00152	-0.00152	0.04326**	0.04326**	0.00569	0.00070	0.00070
	(0.02134)	(0.01832)	(0.01507)	(0.01115)	(0.01407)	(0.02127)	(0.01835)	(0.01519)	(0.01118)	(0.01415)
HI^2	-0.00001*	-0.00001*	-0.00000	0.00000	0.00000	-0.00001*	-0.00001**	-0.00000	0.00000	0.00000
	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)
Hot Days >33°C	0.04566*	0.04566*	-0.01481	-0.01249	-0.01249	0.04497*	0.04497*	-0.00930	-0.00638	-0.00638
	(0.02584)	(0.02598)	(0.02647)	(0.02463)	(0.02380)	(0.02464)	(0.02453)	(0.02498)	(0.02327)	(0.02267)
August Heatwaves	-0.39301	-0.39301	-0.13802	-0.15286	-0.15286	-0.58608**	-0.58608**	-0.51638*	-0.58060**	-0.58060**
	(0.24445)	(0.26396)	(0.25601)	(0.21915)	(0.24272)	(0.26853)	(0.27860)	(0.28019)	(0.23541)	(0.26712)
GSP	-0.00187	-0.00187	0.00443	0.00228	0.00228	-0.00320	-0.00320	0.00345	0.00099	0.00099
	(0.00907)	(0.00932)	(0.00696)	(0.00621)	(0.00654)	(0.00897)	(0.00913)	(0.00688)	(0.00617)	(0.00654)
GSP^2	0.00001	0.00001	-0.00000	0.00000	0.00000	0.00001	0.00001	-0.00000	0.00000	0.00000
	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)
R-squared	0.75771	0.75771	0.42452	0.28105	0.28105	0.75901	0.75901	0.42664	0.28402	0.28402
R-squared (within)	0.08080	0.08080	0.00476	0.02692	0.02692	0.08574	0.08574	0.00843	0.03093	0.03093
Observations	491	491	905	1056	1056	491	491	905	1056	1056

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Panel fixed effects specifications: (1) wine FE with wine clustering; (2) wine FE with winery clustering; (3) winery FE with winery clustering; (4) comune FE with comune clustering; (5) comune FE with winery clustering. HI denotes the Hugin Index. GSP denotes growing season precipitation.

B.4 Price Model – Comparing 97.5th-percentile vs. 99th-percentile Total Season Heatwaves

$$\ln(Price)_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \gamma_1 HD_{35,c,t} + \gamma_2 HW_{tot,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6f')$$

Table B.4: Price Model – Comparing 97.5th-percentile vs. 99th-percentile Total Season Heatwaves

	Price Model (97.5th-percentile)					Price Model (99th-percentile)				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
HI	0.00313***	0.00313***	-0.00211	-0.00092	-0.00092	0.00378***	0.00378***	-0.00166	-0.00075	-0.00075
	(0.00110)	(0.00114)	(0.00244)	(0.00240)	(0.00258)	(0.00116)	(0.00115)	(0.00242)	(0.00245)	(0.00258)
HI^2	-0.00000***	-0.00000**	0.00000	0.00000	0.00000	-0.00000***	-0.00000***	0.00000	0.00000	0.00000
	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)
Hot Days >35°C	0.00892**	0.00892**	-0.00113	-0.00489	-0.00489	0.01217***	0.01217***	0.00349	-0.00322	-0.00322
	(0.00434)	(0.00425)	(0.01205)	(0.01097)	(0.01135)	(0.00430)	(0.00437)	(0.01079)	(0.01035)	(0.01075)
Total Season Heatwaves	-0.02643*	-0.02643*	-0.01096	-0.00273	-0.00273	-0.07329***	-0.07329***	-0.06874	-0.02333	-0.02333
	(0.01481)	(0.01420)	(0.03253)	(0.03275)	(0.03403)	(0.02050)	(0.01930)	(0.04355)	(0.04430)	(0.04633)
GSP	-0.00022	-0.00022	0.00220	0.00330**	0.00330**	-0.00010	-0.00010	0.00227	0.00331**	0.00331**
	(0.00061)	(0.00063)	(0.00167)	(0.00152)	(0.00155)	(0.00059)	(0.00061)	(0.00169)	(0.00152)	(0.00155)
GSP^2	0.00000	0.00000	-0.00000	-0.00000**	-0.00000**	0.00000	0.00000	-0.00000	-0.00000**	-0.00000**
	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)	(0.00000)
R-squared	0.97219	0.97219	0.59531	0.40737	0.40737	0.97277	0.97277	0.59632	0.40750	0.40750
R-squared (within)	0.10199	0.10199	0.04497	0.01885	0.01885	0.12067	0.12067	0.04736	0.01908	0.01908
Observations	453	453	827	972	972	453	453	827	972	972

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Panel fixed effects specifications: (1) wine FE with wine clustering; (2) wine FE with winery clustering; (3) winery FE with winery clustering; (4) comune FE with comune clustering; (5) comune FE with winery clustering. HI denotes the Hugin Index. GSP denotes growing season precipitation. Total season heatwaves are calculated from May to September.

B.5 Temporal Windows – Differential Seasonal Sensitivity

$$Score_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \gamma_1 HD_{33,c,t} + \gamma_2 HW_{m,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6i)$$

Table B.5a: Quality Models by Month

	June Heatwaves (97.5th-percentile)	July Heatwaves (97.5th-percentile)	August Heatwaves (99th-percentile)
HI	0.04353** (0.02133)	0.04113* (0.02174)	0.04326** (0.02127)
HI^2	-0.00001* (0.00000)	-0.00001* (0.00000)	-0.00001* (0.00000)
Hot Days >33°C	0.04456* (0.02505)	0.03731 (0.02527)	0.04497* (0.02464)
Heatwaves (Month)	-0.86671 (0.59563)	0.00946 (0.24961)	-0.58608** (0.26853)
GSP	-0.00153 (0.00918)	-0.00259 (0.00946)	-0.00320 (0.00897)
GSP^2	0.00001 (0.00001)	0.00001 (0.00001)	0.00001 (0.00001)
R-squared (within)	0.08119	0.07637	0.08574
Observations	491	491	491

Notes: All models in this table employ wine-level fixed effects, with standard errors clustered at the wine level. Standard errors are reported in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. May and September models were dropped entirely due to zero variance. June and July rely on the 97.5th-percentile threshold as 99th-percentile events yielded insufficient empirical variance during those months.

$$\ln(Price)_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \gamma_1 HD_{35,c,t} + \gamma_2 HW_{m,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6j)$$

Table B.5b: Price Models by Month vs. Season Total

	June Heatwaves (97.5th-percentile)	July Heatwaves (97.5th-percentile)	Total Season Heatwaves (99%)
HI	0.00317*** (0.00117)	0.00277** (0.00108)	0.00378*** (0.00116)
HI^2	-0.00000** (0.00000)	-0.00000** (0.00000)	-0.00000*** (0.00000)
Hot Days >35°C	0.00773* (0.00432)	0.00573 (0.00423)	0.01217*** (0.00430)
Heatwave (Month / Total)	-0.11213* (0.06693)	-0.00214 (0.01629)	-0.07329*** (0.02050)
GSP	-0.00005 (0.00061)	-0.00021 (0.00062)	-0.00010 (0.00059)
GSP^2	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
R-squared	0.97231	0.97187	0.97277
R-squared (within)	0.10559	0.09146	0.12067
Observations	453	453	453

Notes: All models employ wine-level fixed effects with standard errors clustered at the wine level. Standard errors are reported in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. May and September models were dropped entirely due to zero variance. June and July rely on the 97.5th-percentile threshold as 99th-percentile events yielded insufficient empirical variance during those months.

Appendix C: Sub-Group Analysis Robustness

C.1 Quality Model Analysis by Macro-Terroir and GI Designation

$$Score_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \gamma_1 HD_{33,c,t} + \gamma_2 HW_{aug,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6d'')$$

Table C.1: Quality Model Sub-Group Analysis by Macro-Terroir and GI Designation

	Coastal					Inland				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
HI	0.00937 (0.03291)	0.00937 (0.02995)	-0.00349 (0.02747)	-0.03277 (0.02342)	-0.03277 (0.02648)	0.05497* (0.02989)	0.05497** (0.02673)	0.00842 (0.02131)	0.00568 (0.01494)	0.00568 (0.01900)
HI^2	-0.00000 (0.00001)	-0.00000 (0.00001)	0.00000 (0.00001)	0.00001 (0.00001)	0.00001 (0.00001)	-0.00001 (0.00001)	-0.00001* (0.00001)	-0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000 (0.00000)
Hot Days > 33°C	0.01228 (0.05158)	0.01228 (0.05436)	-0.02231 (0.04035)	-0.02772 (0.04373)	-0.02772 (0.03988)	0.05491* (0.03226)	0.05491* (0.03045)	-0.01287 (0.03501)	-0.00607 (0.03037)	-0.00607 (0.03010)
Aug Heatwaves (99th-percentile)	-0.78768** (0.38946)	-0.78768** (0.38706)	-0.04350 (0.44881)	-0.20210 (0.40130)	-0.20210 (0.46158)	-0.48244 (0.39553)	-0.48244 (0.44961)	-0.64146 (0.38805)	-0.64207** (0.31084)	-0.64207* (0.34752)
GSP	0.02607 (0.01737)	0.02607 (0.01859)	0.02695** (0.01319)	0.01943 (0.01219)	0.01943 (0.01300)	-0.00923 (0.01493)	-0.00923 (0.01594)	-0.00250 (0.01176)	-0.00140 (0.01098)	-0.00140 (0.01140)
GSP^2	-0.00004 (0.00003)	-0.00004 (0.00003)	-0.00004* (0.00002)	-0.00003 (0.00002)	-0.00003 (0.00002)	0.00002 (0.00002)	0.00002 (0.00002)	0.00000 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)
R-squared	0.78448	0.78448	0.51575	0.35294	0.35294	0.76063	0.76063	0.39409	0.25599	0.25599
R-squared (within)	0.11033	0.11033	0.01808	0.02714	0.02714	0.12388	0.12388	0.01026	0.03424	0.03424
Observations	191	191	330	369	369	264	264	498	604	604
	PDO					PGI				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
HI	0.02148 (0.03011)	0.02148 (0.03034)	0.00335 (0.02287)	0.00104 (0.01483)	0.00104 (0.02156)	0.08241*** (0.02965)	0.08241*** (0.02901)	0.03301 (0.02966)	0.02217 (0.02528)	0.02217 (0.02992)
HI^2	-0.00000 (0.00001)	-0.00000 (0.00001)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	-0.00002** (0.00001)	-0.00002*** (0.00001)	-0.00001 (0.00001)	-0.00000 (0.00001)	-0.00000 (0.00001)
Hot Days > 33°C	-0.00235 (0.03211)	-0.00235 (0.03480)	-0.10213*** (0.03404)	-0.06764** (0.02896)	-0.06764** (0.02884)	0.09949*** (0.03674)	0.09949*** (0.04030)	0.06189* (0.03158)	0.06361* (0.03379)	0.06361* (0.03416)
Aug Heatwaves (99th-percentile)	-0.03498 (0.42427)	-0.03498 (0.42289)	0.53489 (0.39680)	0.21202 (0.32737)	0.21202 (0.36431)	-1.06618*** (0.32206)	-1.06618*** (0.33653)	-1.48035*** (0.38691)	-1.38451*** (0.36714)	-1.38451*** (0.41773)
GSP	-0.01197 (0.01104)	-0.01197 (0.01122)	-0.00148 (0.00790)	0.00147 (0.00671)	0.00147 (0.00750)	0.00906 (0.01605)	0.00906 (0.01697)	-0.00821 (0.01288)	-0.00860 (0.01275)	-0.00860 (0.01360)
GSP^2	0.00002 (0.00001)	0.00002 (0.00001)	0.00001 (0.00001)	0.00000 (0.00001)	0.00000 (0.00001)	-0.00000 (0.00002)	-0.00000 (0.00002)	0.00001 (0.00002)	0.00001 (0.00002)	0.00001 (0.00002)
R-squared	0.78726	0.78726	0.51408	0.40627	0.40627	0.70140	0.70140	0.50232	0.39776	0.39776
R-squared (within)	0.10925	0.10925	0.05502	0.07144	0.07144	0.13597	0.13597	0.05378	0.03793	0.03793
Observations	267	267	454	570	570	224	224	383	456	456

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Panel fixed effects specifications: (1) wine FE with wine clustering; (2) wine FE with winery clustering; (3) winery FE with winery clustering; (4) comune FE with comune clustering; (5) comune FE with winery clustering. GSP is growing season precipitation. Coastal provinces: Grosseto, Livorno, Pisa. Inland provinces: Florence, Siena, Arezzo. HI denotes the Hugin Index. GSP denotes growing season precipitation.

C.2 Price Model Analysis by Macro-Terroir and GI Designation

$$\ln(Price)_{ijc,t} = \alpha + \beta_1 HI_{c,t} + \beta_2 HI_{c,t}^2 + \gamma_1 HD_{35,c,t} + \gamma_2 HW_{tot,c,t} + \theta_1 GSP_{c,t} + \theta_2 GSP_{c,t}^2 + \lambda Vintage_t + \delta_s + \epsilon_{ijc,t} \quad (4.6f'')$$

Table C.2: Price Model Sub-Group Analysis by Macro-Terroir and GI Designation

	Coastal					Inland				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
HI	0.00561** (0.00232)	0.00561** (0.00246)	0.00780 (0.00718)	0.00173 (0.00711)	0.00173 (0.00672)	0.00178* (0.00103)	0.00178* (0.00095)	-0.00343 (0.00300)	-0.00127 (0.00302)	-0.00127 (0.00321)
HI^2	-0.00000** (0.00000)	-0.00000** (0.00000)	-0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
Hot Days > 35°C	0.00750 (0.00855)	0.00750 (0.00891)	0.00867 (0.02347)	0.01172 (0.02296)	0.01172 (0.02400)	0.01083** (0.00424)	0.01083** (0.00485)	0.00774 (0.01363)	-0.00442 (0.01336)	-0.00442 (0.01360)
Season Heatwaves (99th-percentile)	-0.06585* (0.03703)	-0.06585* (0.03727)	0.00037 (0.05702)	0.01647 (0.07442)	0.01647 (0.07169)	-0.08316*** (0.02694)	-0.08316*** (0.02919)	-0.14492** (0.07274)	-0.04951 (0.06342)	-0.04951 (0.06783)
GSP	0.00019 (0.00117)	0.00019 (0.00118)	-0.00094 (0.00244)	0.00062 (0.00219)	0.00062 (0.00235)	-0.00009 (0.00094)	-0.00009 (0.00089)	0.00365 (0.00220)	0.00449* (0.00241)	0.00449* (0.00229)
GSP^2	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	-0.00001** (0.00000)	-0.00001** (0.00000)	-0.00001** (0.00000)
R-squared	0.97126	0.97126	0.58343	0.43098	0.43098	0.98046	0.98046	0.62483	0.39675	0.39675
R-squared (within)	0.13599	0.13599	0.05146	0.03380	0.03380	0.10909	0.10909	0.06918	0.01915	0.01915
Observations	175	175	297	336	336	248	248	465	565	565
	PDO					PGI				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
HI	0.00201 (0.00142)	0.00201 (0.00139)	-0.00278 (0.00319)	-0.00035 (0.00250)	-0.00035 (0.00314)	0.00532*** (0.00162)	0.00532*** (0.00183)	0.00548 (0.00540)	0.00656 (0.00571)	0.00656 (0.00574)
HI^2	-0.00000 (0.00000)	-0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)	-0.00000*** (0.00000)	-0.00000** (0.00000)	-0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000 (0.00000)
Hot Days > 35°C	0.01023* (0.00598)	0.01023* (0.00596)	0.00183 (0.01056)	-0.00254 (0.01001)	-0.00254 (0.01125)	0.01429** (0.00566)	0.01429** (0.00566)	0.00613 (0.01819)	0.01125 (0.01791)	0.01125 (0.01807)
Season Heatwaves (99th-percentile)	-0.06728*** (0.02558)	-0.06728*** (0.02270)	-0.03233 (0.04889)	-0.01045 (0.05009)	-0.01045 (0.05149)	-0.07063** (0.03023)	-0.07063** (0.02835)	-0.10075 (0.08045)	-0.13306 (0.08104)	-0.13306 (0.08589)
GSP	-0.00079 (0.00055)	-0.00079 (0.00059)	0.00290 (0.00224)	0.00306* (0.00176)	0.00306 (0.00186)	0.00126 (0.00111)	0.00126 (0.00121)	-0.00104 (0.00287)	0.00051 (0.00293)	0.00051 (0.00302)
GSP^2	0.00000 (0.00000)	0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000* (0.00000)	-0.00000* (0.00000)	-0.00000 (0.00000)	-0.00000 (0.00000)	0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000 (0.00000)
R-squared	0.96207	0.96207	0.59622	0.44934	0.44934	0.97989	0.97989	0.63941	0.53380	0.53380
R-squared (within)	0.23903	0.23903	0.04134	0.02053	0.02053	0.11775	0.11775	0.05107	0.04548	0.04548
Observations	245	245	425	532	532	208	208	346	413	413

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Panel fixed effects specifications: (1) wine FE with wine clustering; (2) wine FE with winery clustering; (3) winery FE with winery clustering; (4) comune FE with comune clustering; (5) comune FE with winery clustering. GSP is growing season precipitation. Coastal provinces: Grosseto, Livorno, Pisa. Inland provinces: Florence, Siena, Arezzo. HI denotes the Huglin Index. GSP denotes growing season precipitation. Season Heatwaves denotes the 99th-percentile total season heatwaves from May to September.